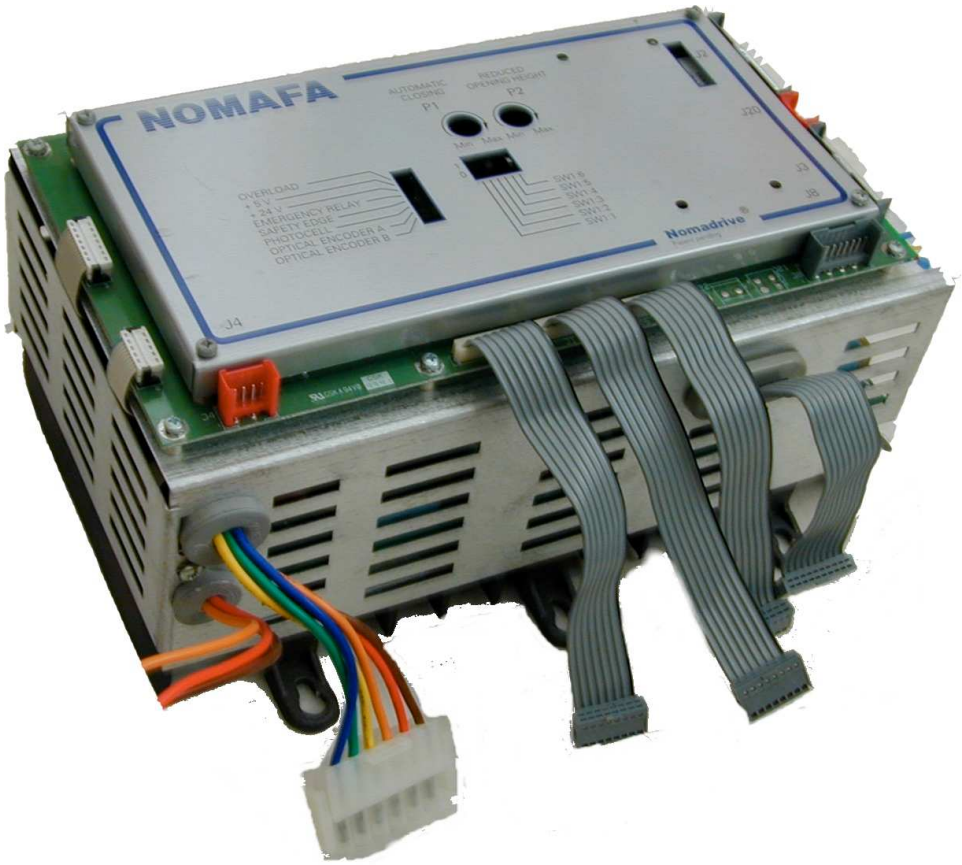


ACS70 or NomaDrive Manual

Directions for use and trouble shooting for Resellers in Asia

Albany Door Systems (Asia)
Regional Office Malaysia
version 2006-1
Th. Matter

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Attention: Be aware of the warning on pages 38 & 39 about terminal nr. Changes

General description:

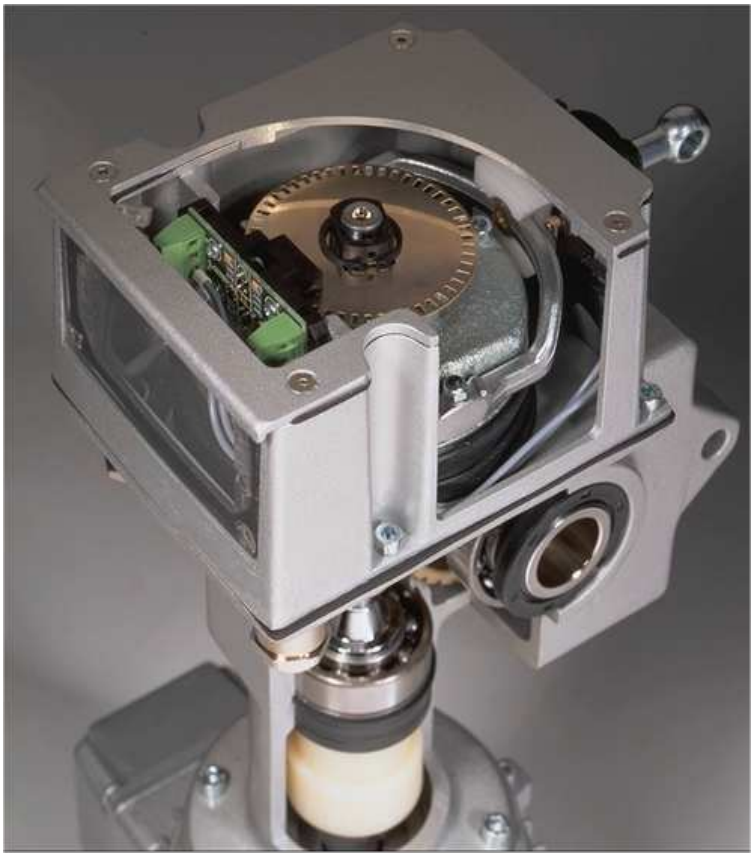


CPU

Drive-unit



Encoder housing



Theory block 1 :

The NomaDrive system comprises:

- **CPU**
- **Drive unit**
- **Peripherals**

CPU

Comprises : Motor card / Computer card / Display with keys

- **Converts AC power into DC power**
- **Distributes the DC power frequency controlled to the drive unit**
- **Monitors the position of the door (counting pulses from encoder)**
- **Maintains a fail-safe system**
- **Organizes the activators with their logic's**
- **Power supply for peripherals / activators**
- **Interfaces with external peripherals**
- **Dynamical DC-braking system**

Terminal card

- **Inputs for power**
- **Inputs for peripherals and activator signals**
- **Outputs to Motor**
- **Outputs to traffic lights**
- **Outputs to status relays**

Drive Unit

- **Motor (AC squirrel cage type, 3-phase)**
- **Hypoid worm gear transmission**
- **Optical encoder**
- **Electro-magnetic parking brake**
- **Brake release handle**

Housing

- **LCD-Display + LED's for error messages and Power ON**
- **Main switch**
- **Emergency stop switch**
- **Open / Close button**
- **Terminal card**
- **CPU**

- **Circuit breaker**

Theory block 2:

1. **The Mains supply of AC power comes into the CPU and is filtered from spikes and other interferences.**

2. **Via an Emergency stop relay contact it reaches a rectifier bridge which converts the AC power into (intermittend) DC power. (both halves of the sinus)**

3. **The capacitor bank is fed via a resistor (which limits the loading current during start-up) which is bridged with a load relay contact after start-up. This capacitor bank is the storage medium for the power to the motor.**

4. **The stored DC power is distributed via an IGBT module (power transistors in an integrated circuit pack) to the motor in 3 separated channels (phases) to the U, V, W lines.**

5. **The speed of the motor is controlled by the duration, frequency and firing order of the power pulses send from the capacitor bank to the motor via this IGBT module.**

6. **The frequency, pulse duration and firing order of the IGBT module controlled by the computer card in the CPU. It generates all the needed Signals with a micro-processor and sends the signals to the IGBT module via gates a, b, c, d, e and f.**

7. **The speed is dependent of the parameter setting in the CPU's memory chips (EPROM and EEPROM), and the position of the door blade in the door. (Open or closed or in between)**

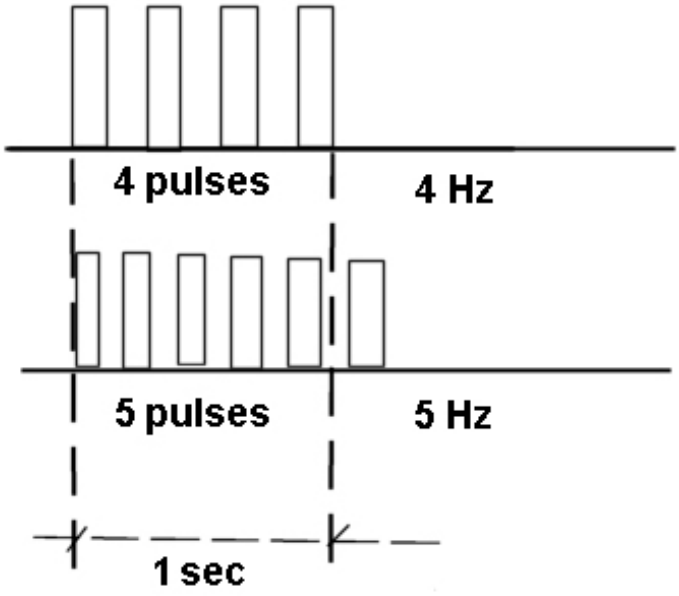
8. **The activators and the safety devices like safety edge and photocells Etc. start the process of motion of the door blade and limit the process as well.**

9. **The parameters for the specific door can be programmed with a hand-held programming device: "Service Instrument"**

10. **The default parameters are stored in the EPROM. The variable and door specific parameters are stored in the EEPROM, and are sometimes amended during the operation of the door.**

Note: **I**nsulated **G**ate **B**ipolar **T**ransistor (IGBT) a MOS FET development

Frequency = pulses/sec



Electric induction motors are speed dependent of Frequency.

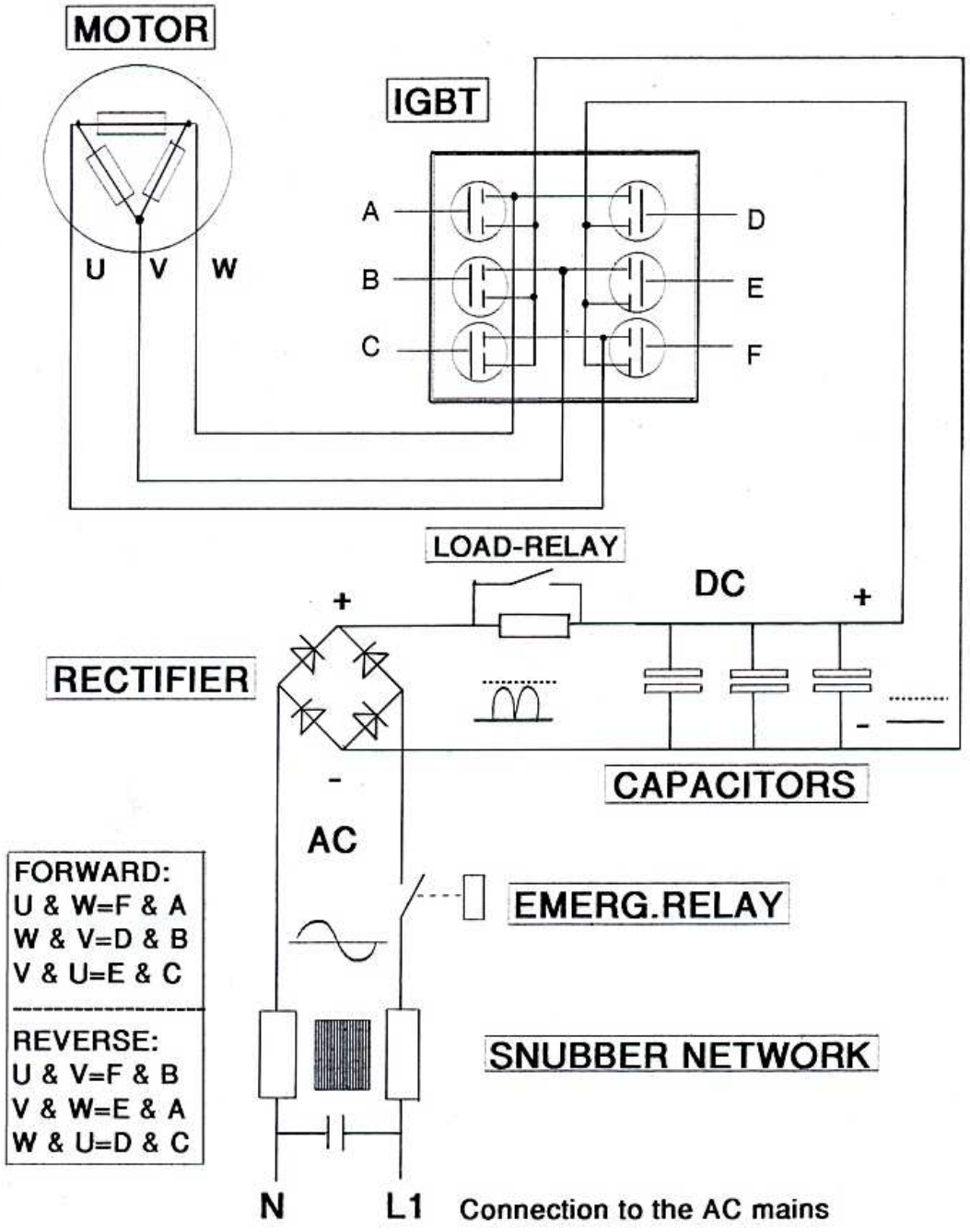
Low frequency = slow speed

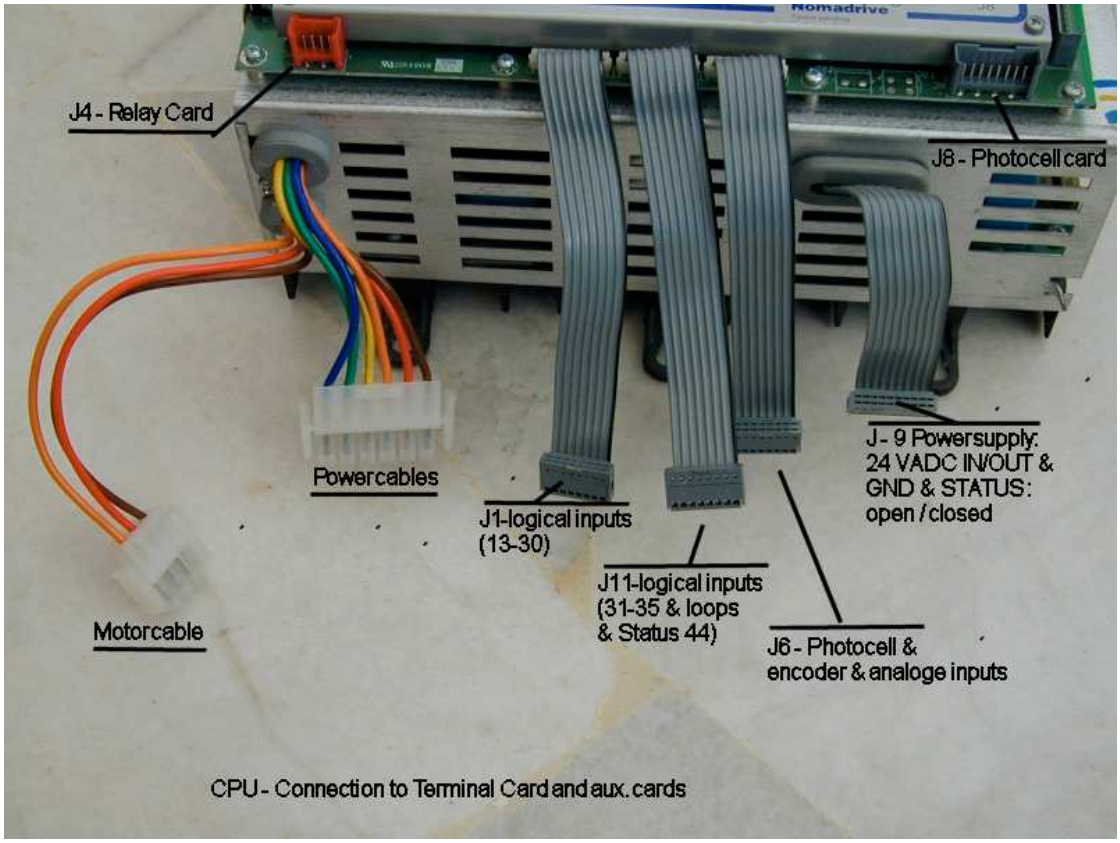
High Frequency = high speed

Stepless controlled speed, varies the frequency on a "gliding scale" upwards and downwards.

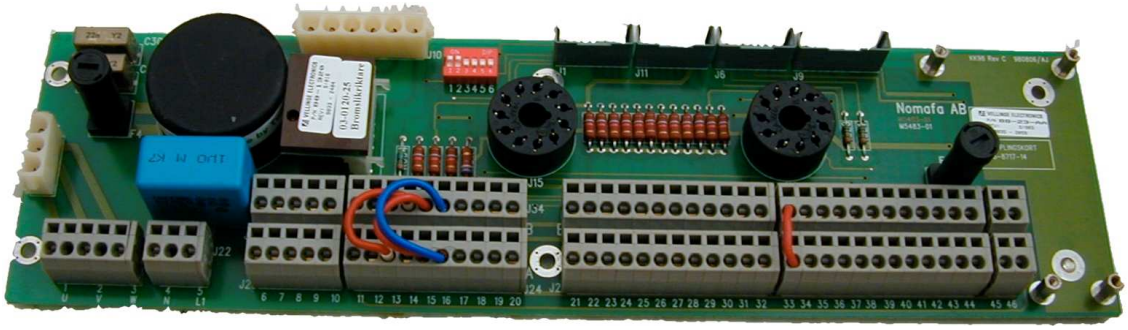
The "flywheel effect" of the mass of the armature causes a smooth motion.

Conversion from AC to DC to Frequency controlled distributed Pulsed DC

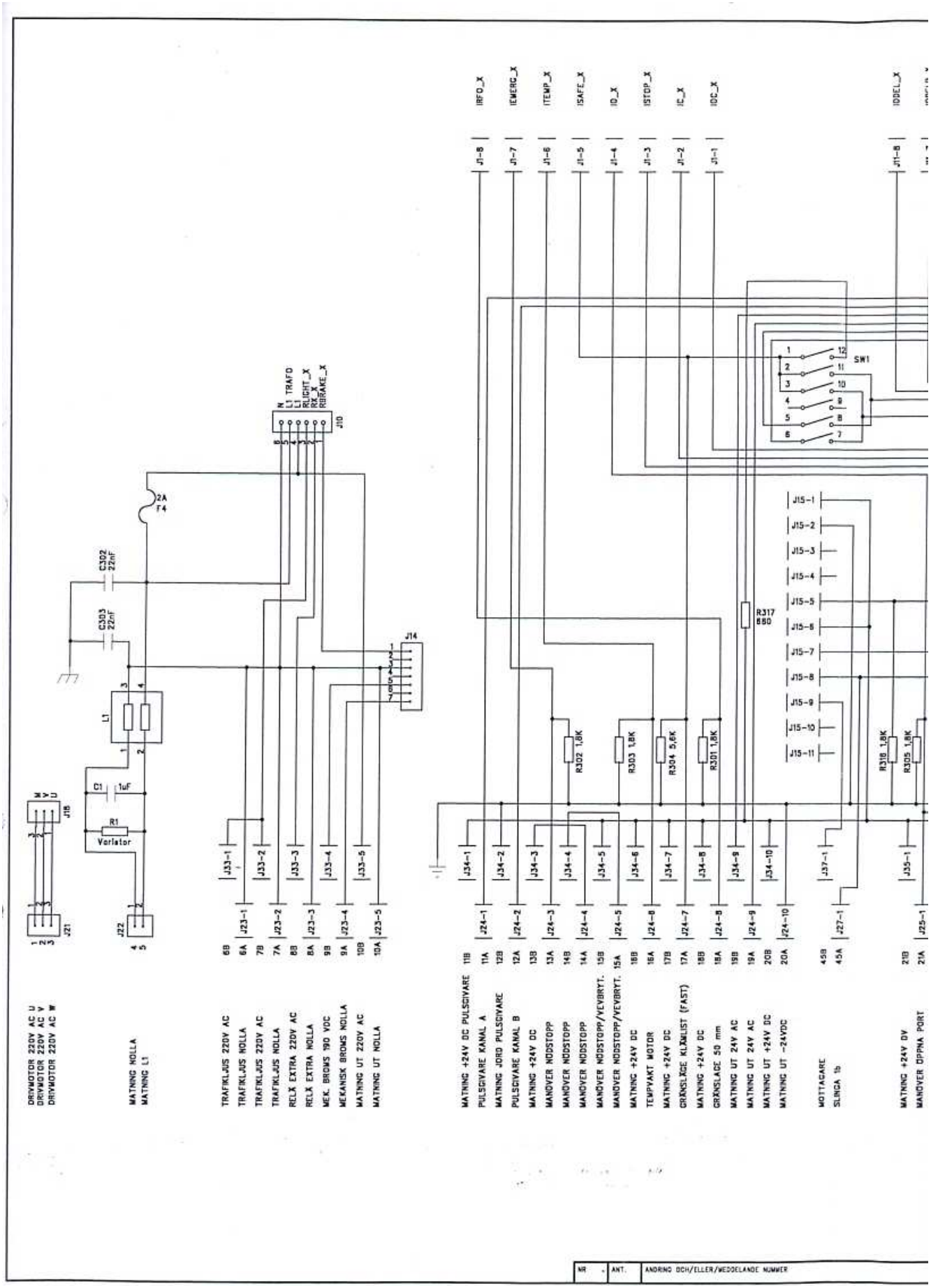


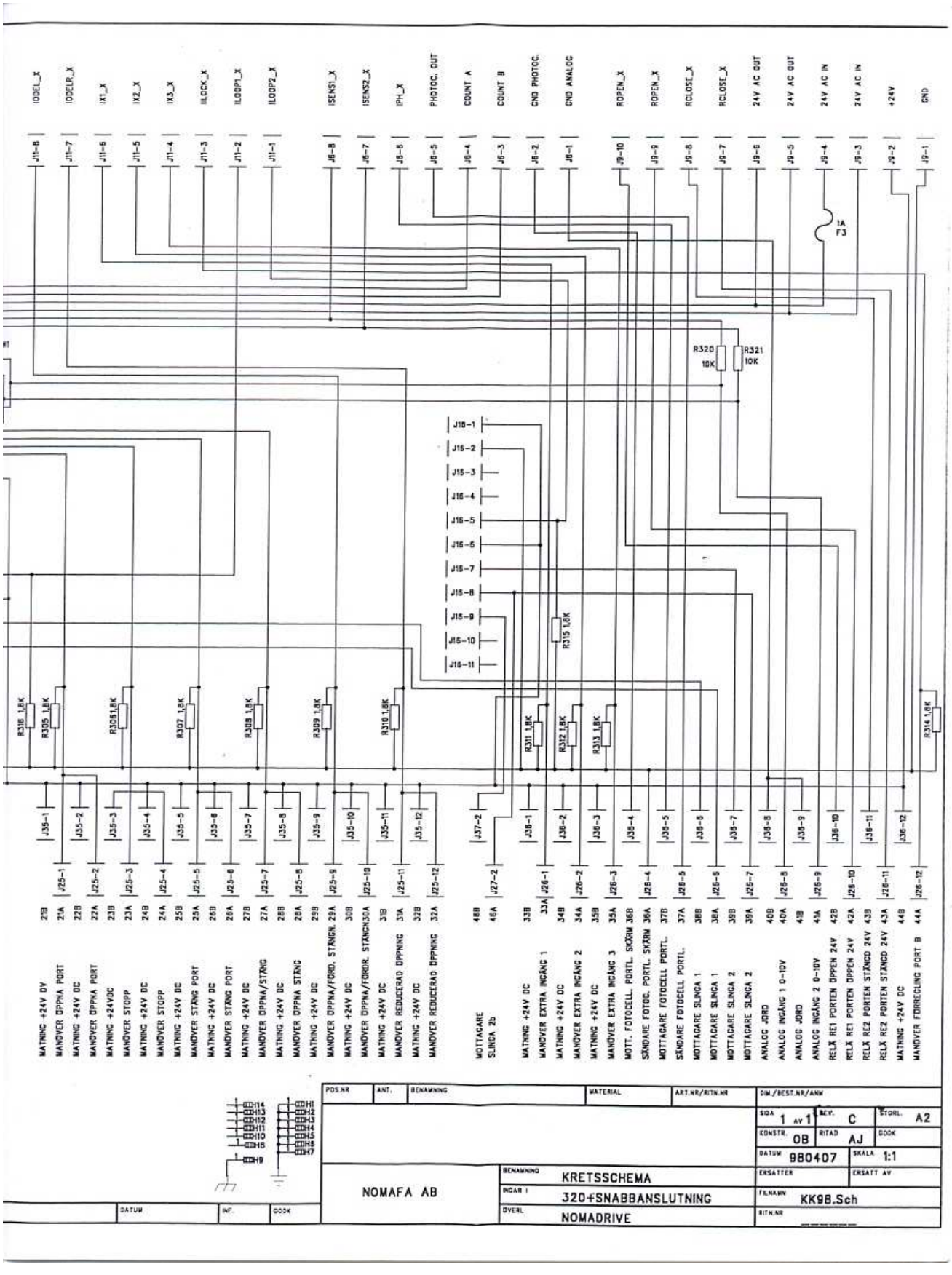


CPU with cable connections to Terminal card

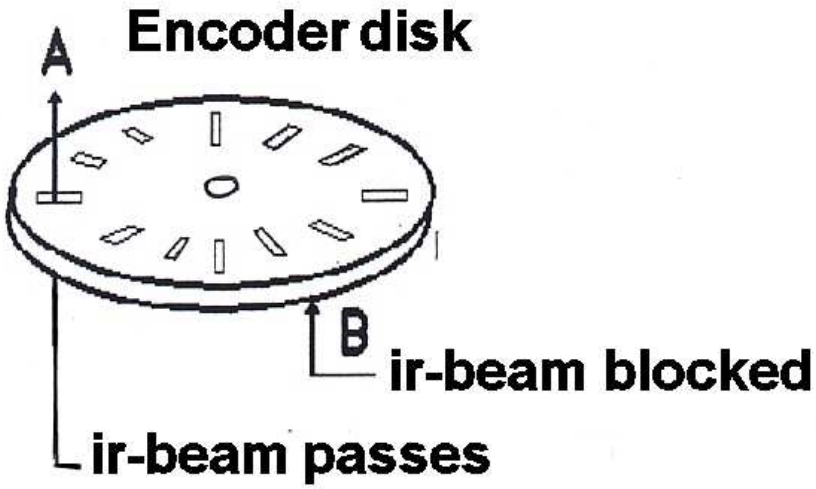


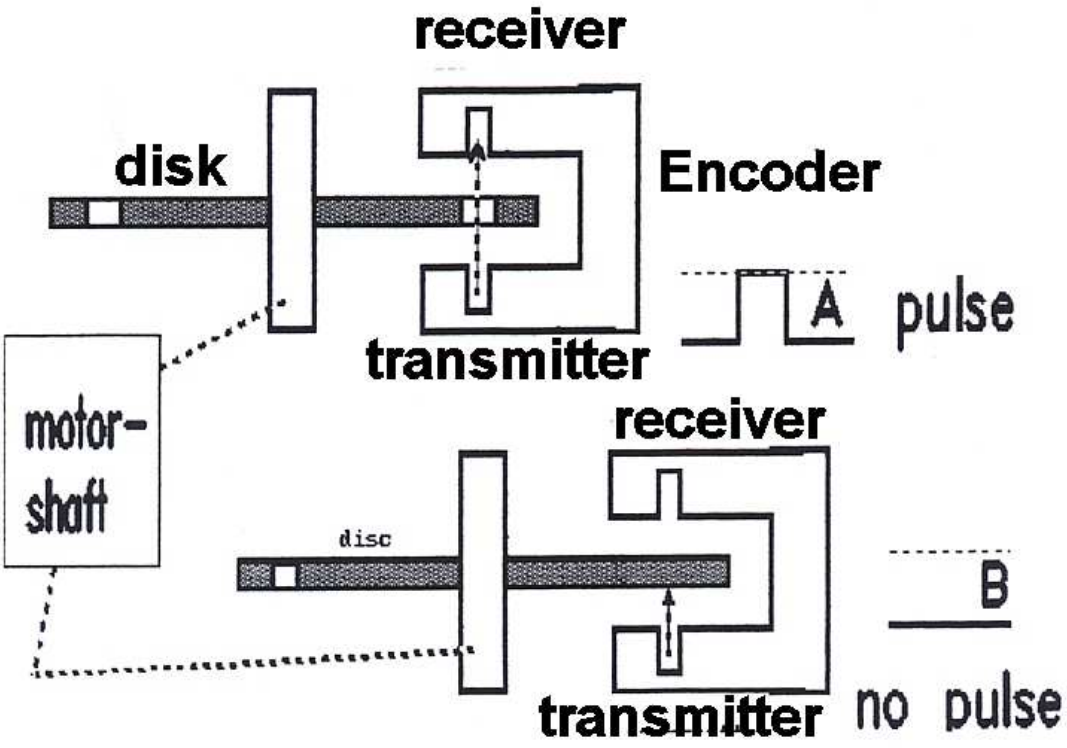
Terminal card with Fuses, terminals and interfacing to the Mains – supply


DRIVMOTOR 220V AC U
DRIVMOTOR 220V AC V
DRIVMOTOR 220V AC W
MATNING NOLLA
MATNING L1
TRAFIKLJUS 220V AC
TRAFIKLJUS NOLLA
TRAFIKLJUS 220V AC
TRAFIKLJUS NOLLA
RELK EXTRA 220V AC
RELK EXTRA NOLLA
MEK. BROMS 190 VDC
MEKANISK BROMS NOLLA
MATNING UT 220V AC
MATNING UT NOLLA
MATNING +24V DC PULSGIVARE
PULSGIVARE KANAL A
MATNING JORD PULSGIVARE
PULSGIVARE KANAL B
MATNING +24V DC
MANÖVER NÖDSTOPP
MANÖVER NÖDSTOPP
MANÖVER NÖDSTOPP/VEVBRYT.
MANÖVER NÖDSTOPP/VEVBRYT.
MATNING +24V DC
TEMPVAKT MOTOR
MATNING +24V DC
GRÄNSLÄGE KLÄMLIST (FAST)
MATNING +24V DC
GRÄNSLÄGE 50 mm
MATNING UT 24V AC
MATNING UT 24V AC
MATNING UT +24V DC
MATNING UT -24VDC
MOTTAGARE
SLINGA 1b
MATNING +24V DV
MANÖVER ÖPPNA PORT
IRFO_X
IEWERG_X
ITEMP_X
ISAFE_X
ID_X
ISTOP_X
IC_X
IDC_X
IDDEL_X
N
L1 TRAFO
L1
LIGHT_X
RELX
RBRAKE_X
J10
2A
F4
C302 22nF
C303 22nF
L1
C1 1uF
R1
Varistor
J14
J18
J21
J22
J23-1
J23-2
J23-3
J23-4
J23-5
J33-1
J33-2
J33-3
J33-4
J33-5
J34-1
J34-2
J34-3
J34-4
J34-5
J34-6
J34-7
J34-8
J34-9
J34-10
J37-1
J35-1
J24-1
J24-2
J24-3
J24-4
J24-5
J24-6
J24-7
J24-8
J24-9
J24-10
J27-1
J25-1
R302 1,8K
R303 1,8K
R304 5,6K
R301 1,8K
R317 680
R316 1,8K
R305 1,8K
SW1
J15-1
J15-2
J15-3
J15-4
J15-5
J15-6
J15-7
J15-8
J15-9
J15-10
J15-11
NR
ANT.
ANDRING OCH/ELLER/WEDDELANDE NUMMER


IODEL_X
IODELR_X
IX1_X
IX2_X
IX3_X
ILOCK_X
ILOOP1_X
ILOOP2_X
ISENS1_X
ISENS2_X
IPH_X
PHOTOC. OUT
COUNT A
COUNT B
GND PHOTOC.
GND ANALOG
ROPEN_X
ROPEN_X
RCLOSE_X
RCLOSE_X
24V AC OUT
24V AC OUT
24V AC IN
24V AC IN
+24V
GND
1A F3
R320 10K
R321 10K
R315 1,8K
R318 1,8K
R305 1,8K
R306 1,8K
R307 1,8K
R308 1,8K
R309 1,8K
R310 1,8K
R311 1,8K
R312 1,8K
R313 1,8K
R314 1,8K
MATNING +24V DV
MANÖVER ÖPPNA PORT
MATNING +24V DC
MANÖVER ÖPPNA PORT
MATNING +24VDC
MANÖVER STOPP
MATNING +24V DC
MANÖVER STOPP
MATNING +24V DC
MANÖVER STÄNG PORT
MATNING +24V DC
MANÖVER STÄNG PORT
MATNING +24V DC
MANÖVER ÖPPNA/STÄNG
MATNING +24V DC
MANÖVER ÖPPNA STÄNG
MATNING +24V DC
MANÖVER ÖPPNA/FÖRD. STÄNGN.
MATNING +24V DC
MANÖVER ÖPPNA/FÖRDR. STÄNGN30A
MATNING +24V DC
MANÖVER REDUCERAD ÖPPNING
MATNING +24V DC
MANÖVER REDUCERAD ÖPPNING
MOTTAGARE
SLINGA 2b
MATNING +24V DC
MANÖVER EXTRA INGÅNG 1
MATNING +24V DC
MANÖVER EXTRA INGÅNG 2
MATNING +24V DC
MANÖVER EXTRA INGÅNG 3
MOTT. FOTOCELL PORTL. SKÄRM
SÄNDARE FOTOC. PORTL. SKÄRM
MOTTAGARE FOTOCELL PORTL.
SÄNDARE FOTOCELL PORTL.
MOTTAGARE SLINGA 1
MOTTAGARE SLINGA 1
MOTTAGARE SLINGA 2
MOTTAGARE SLINGA 2
ANALOG JORD
ANALOG INGÅNG 1 0-10V
ANALOG JORD
ANALOG INGÅNG 2 0-10V
RELÄ RE1 PORTEN ÖPPEN 24V
RELÄ RE1 PORTEN ÖPPEN 24V
RELÄ RE2 PORTEN STÄNGD 24V
MATNING +24V DC
MANÖVER FÖRREGLING PORT B
POS.NR
ANT.
BENÄMNING
MATERIAL
ART.NR/RITN.NR
DIM./BEST.NR/ANM
SIDA 1 AV 1
REV. C
STORL. A2
KONSTR. OB
RITAD AJ
DOK
DATUM 980407
SKALA 1:1
ERSÄTTER
ERSÄTT AV
FILNAMN KK98.Sch
RITN.NR
NOMAFA AB
BENÄMNING KRETSSCHEMA
INGÅR I 320+SNABBANSLUTNING
ÖVERL NOMADRIVE
DATUM
INF.
DOK

NomaDrive Encoder system







Encoder housing on top of transmission (brake and encoder + micro-switch)

THEORY BLOCK 3:

OPTICAL ENCODER:

The encoder disk (with 50 slits) is placed on the motor shaft and runs with the speed of the drive motor.

The disk is passing two separate optic sensors which are counting the passing Infra-red beams through the slits at any one time.

The counted number of pulses (or counts) are passed onto the CPU.
The CPU is able to check the direction that the disk turns.

Each time the safety edge of the door generates a "zero" pulse when hitting the floor (or 50 mm switch in doors with optical safety edge, or lower limit switch in other door models), the counter in the CPU is reset to "0".

By counting from "0" up to the highest number of counts possible by reaching the highest opening height, the CPU is able to be aware at what position the door blade is at any one time during the whole door cycle.

Attention: The number of counts on the LCD display are just "counts" and NOT centimeters or millimeters !!!

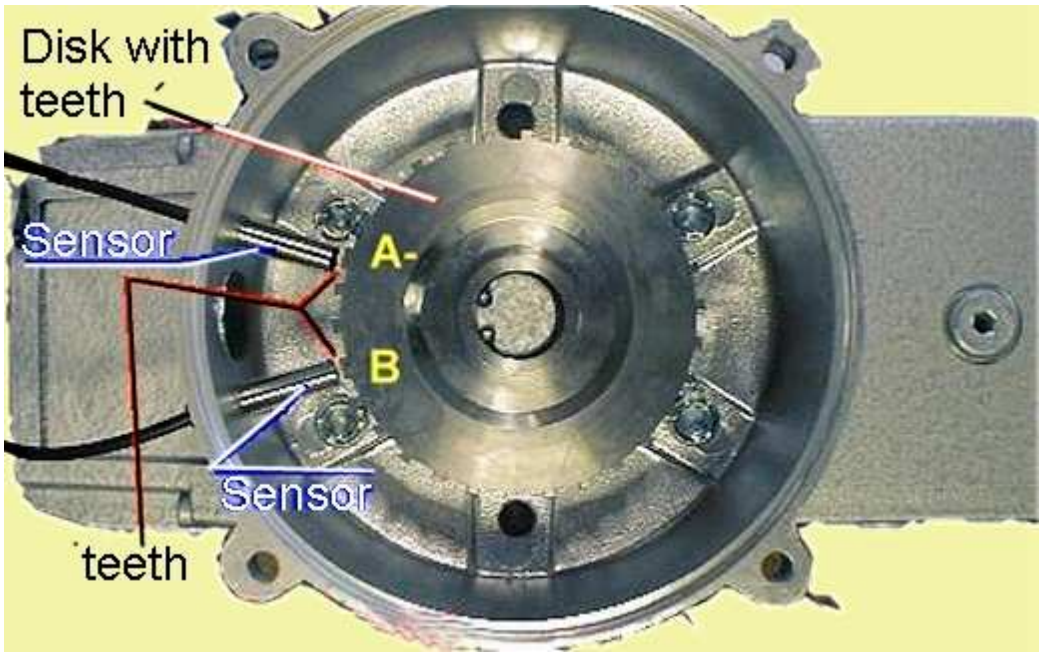
The pulses of the encoder are also visible on the front of the encoder housing : 2 LED's are visible through the window in the housing on top of the transmission.
The CPU has also 2 LED's indicating the pulses from the encoder:
LED's A & B on the front plate (lowest 2 LED's)

IMPORTANT: Always check for proper working order of counter

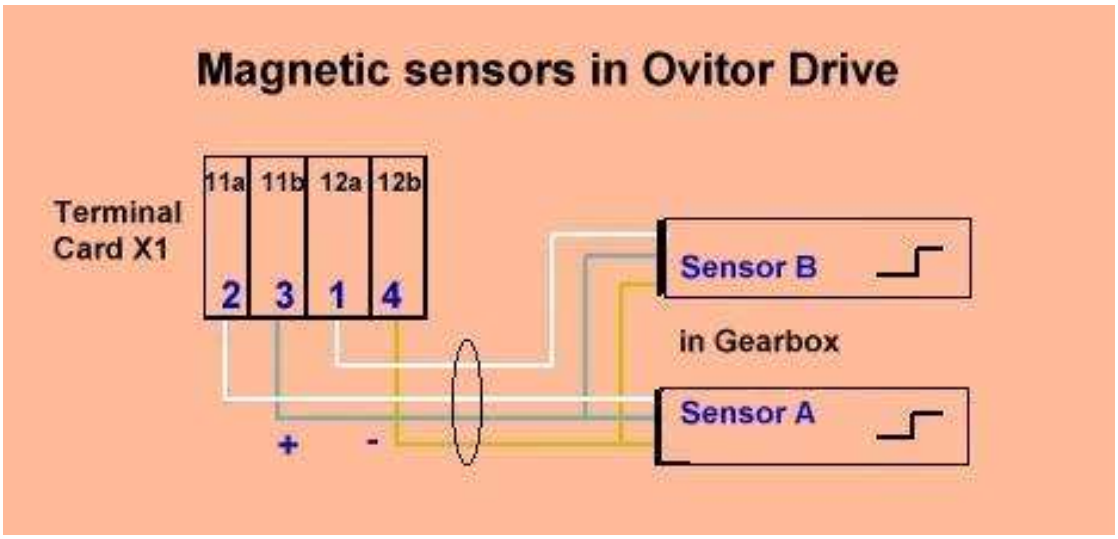
"ZERO POSITION" when safety edge has touched the floor

(or zero switch in other doors has been activated)

Read the latest information about Encoder disks in ACS70 systems, at page 76 & 77

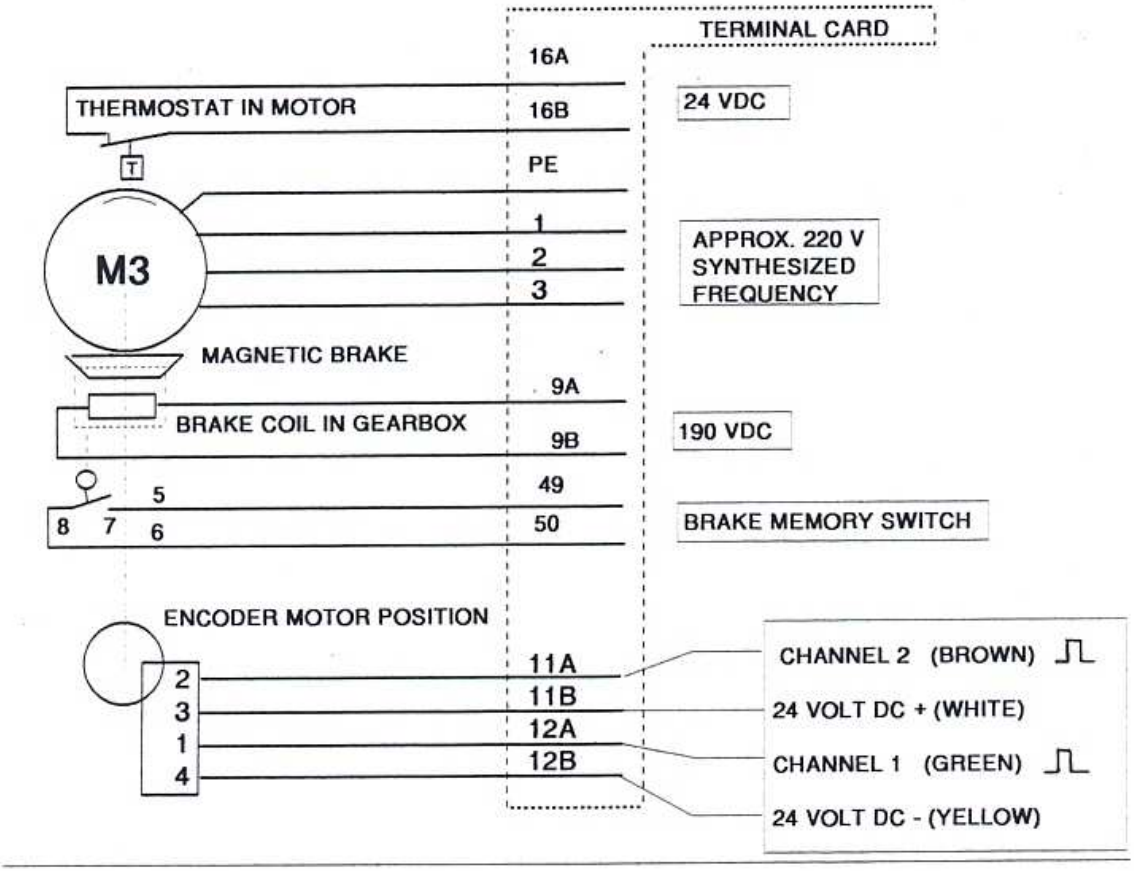


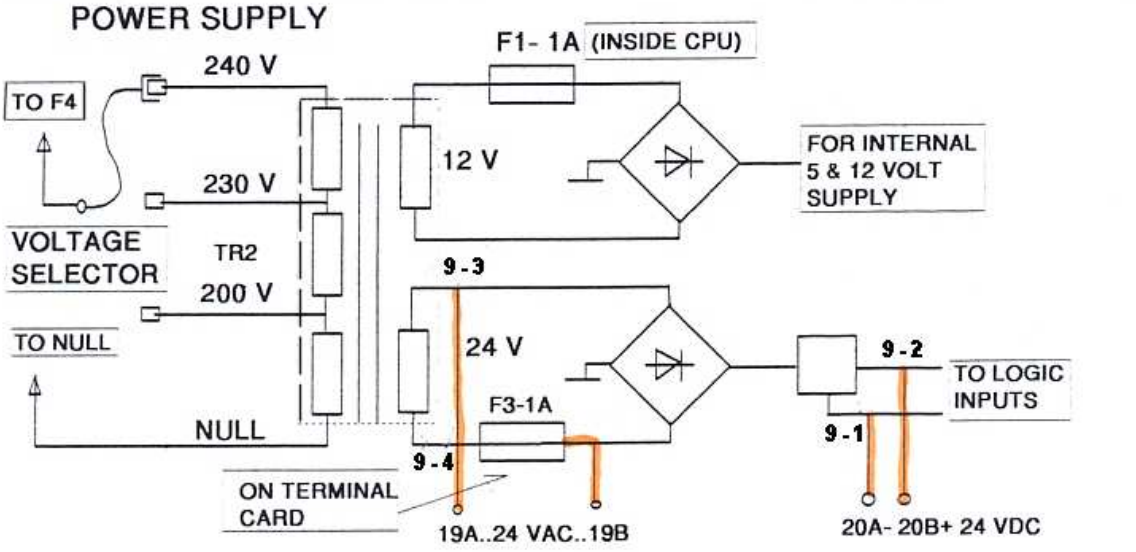
Magnetic sensors in Ovitor drive for the use of NomaDrive system counter in 300 & 700 series doors with Ovitor drive unit.
(air-gap 0.4 mm between sensor and teeth)



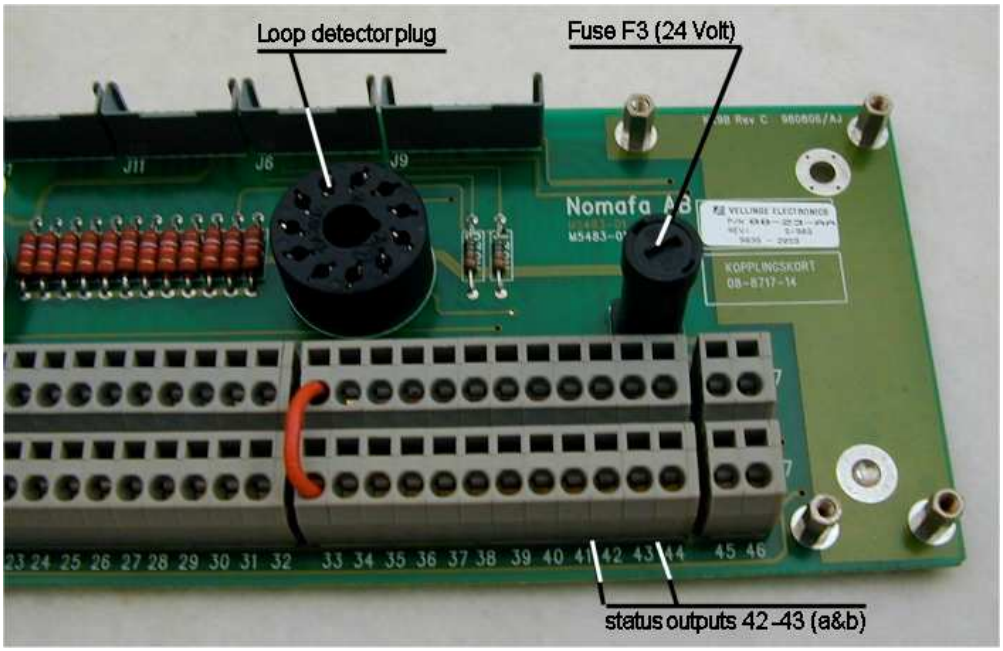
Terminal connections on NomaDrive terminal card for OVITOR drives

NomaDrive motor connections

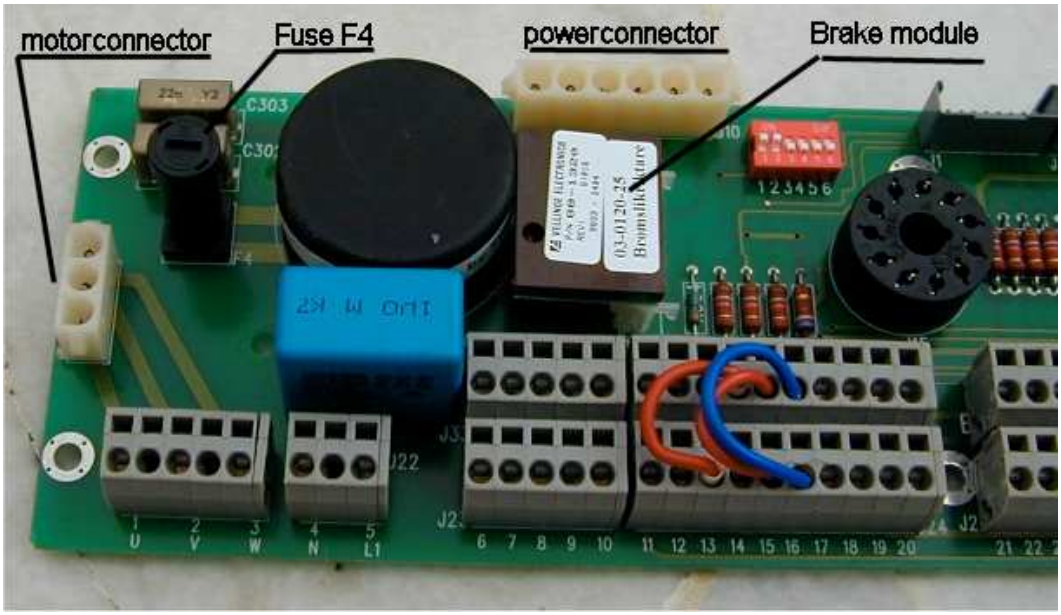




Details of Terminal card, with Fuse and selection Dip-switches and Brake module



Detail Terminal card with loop detector plug, Fuse 3 and status outputs

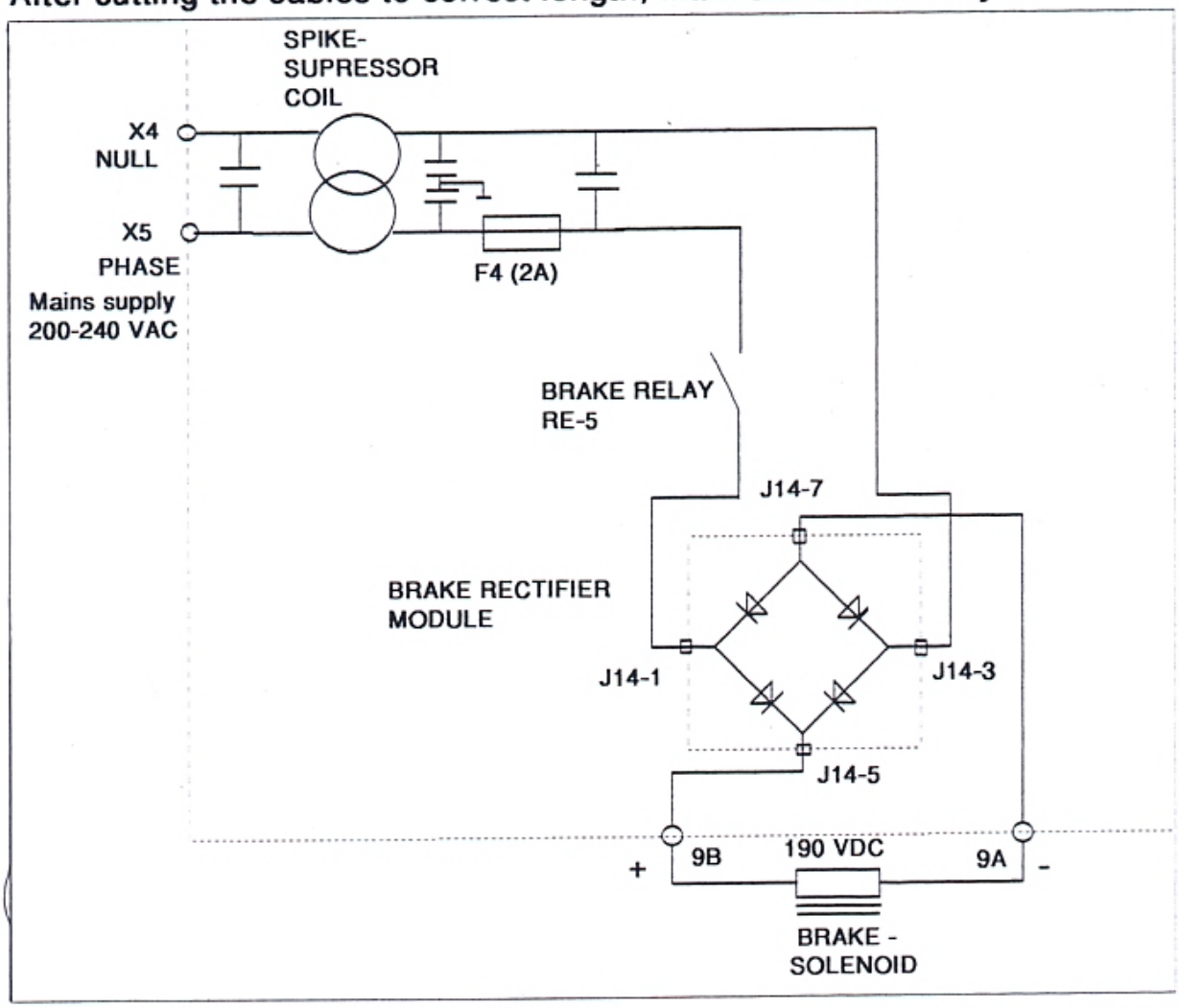


Terminal card detail : Power connections + brake module + Fuse 4

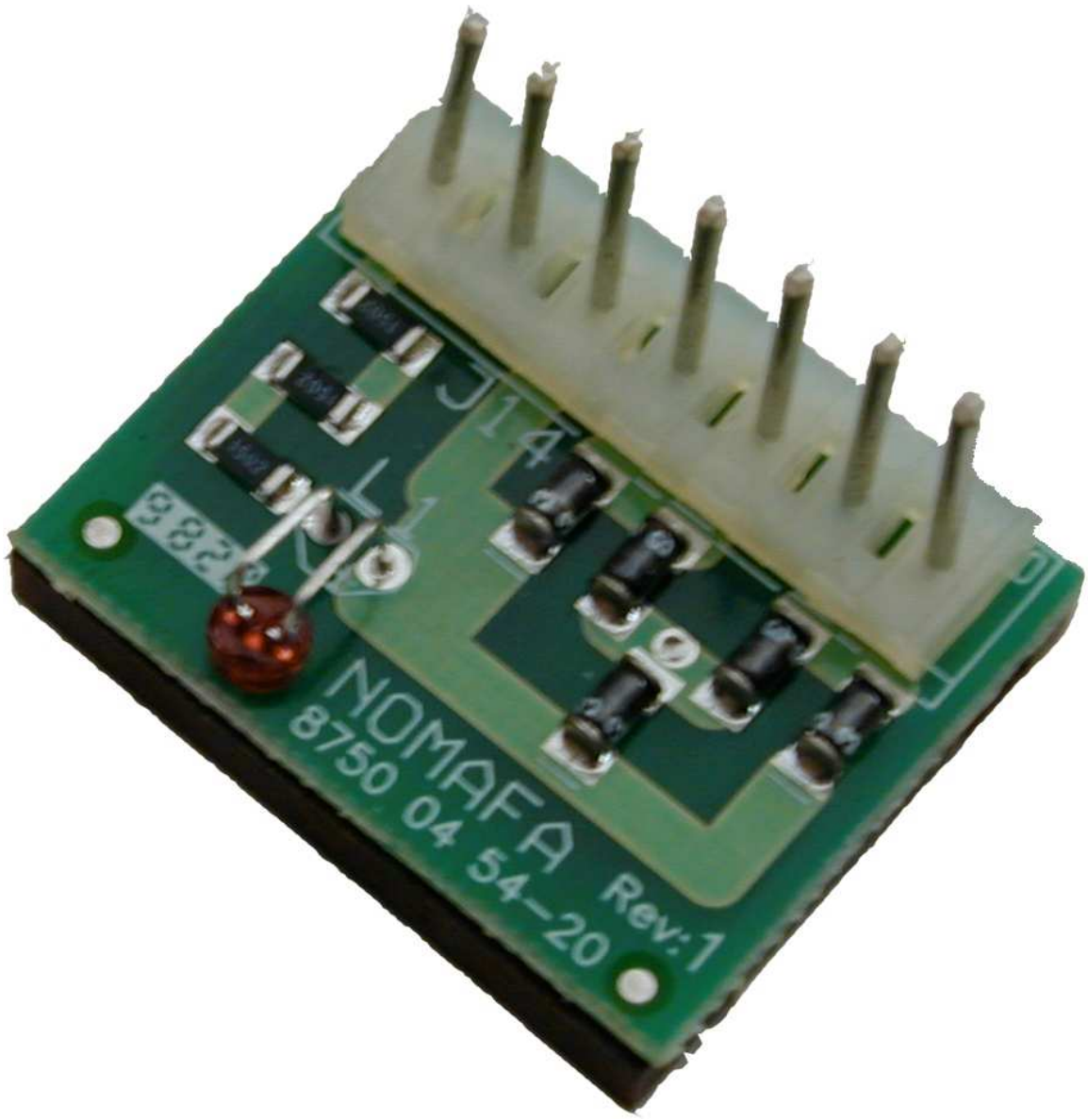
NomaDrive : Brake rectifier circuit

The electric motor of the Nomadrive has 3 cables connected to the Nomadrive Control box : 1. Motor cable marked : 1 - 2 - 3 - PE
2. Electric Brake cable : 9A - 9B
3. Thermostat cable : 16A - 16B

After cutting the cables to correct length, mark cables correctly !



WARNING : CONNECT ONLY THE EL.BRAKE CABLE TO 9A + 9B, NEVER CONNECT THE THERMOSTAT CABLE TO 9A + 9B, SINCE THAT WILL RESULT IN IMMEDIATE BLOWING OF FUSE F4, CAUSED BY DESTROYED BRAKE RECTIFIER MODULE AT START HOLD TO RUN

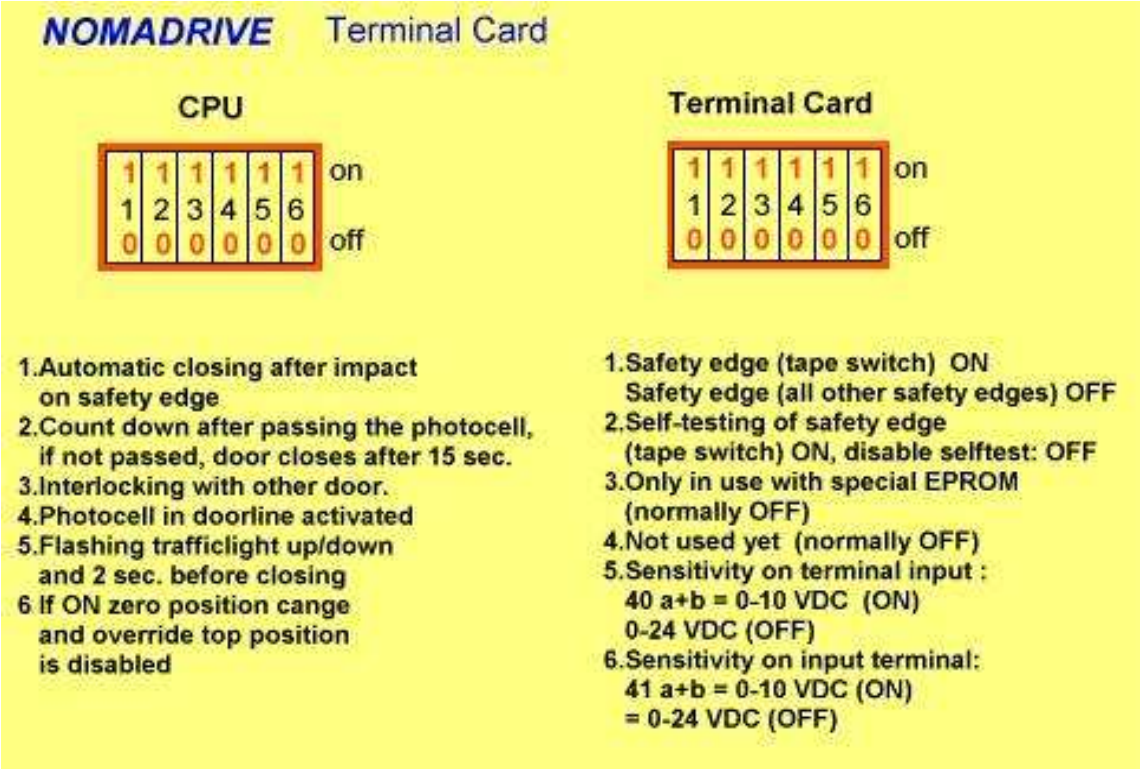


Brake module, located on the Terminal card with LED for indication of being used

The brake module is only used during the “parking brake” cycle of the door.

(The “parking brake cycle” is when the door is standing still, no motion at all.)

All other braking is carried out by the Electronic DC-braking mode of the CPU.



The CPU and the Terminal card have both DIP switches (red switch blocks):

On the **CPU** it regulates following functions :

- activators and closing modes.
- the interlock
- traffic lights mode
- In Hold to run mode: Top position override and zero position change

On the **Terminal card** the following functions are regulated:

- safety edge mode switching
- self-testing of safety edge function
- sesitivity setting of inputs 40 and 41 (between 10 VDC and 24 VDC)

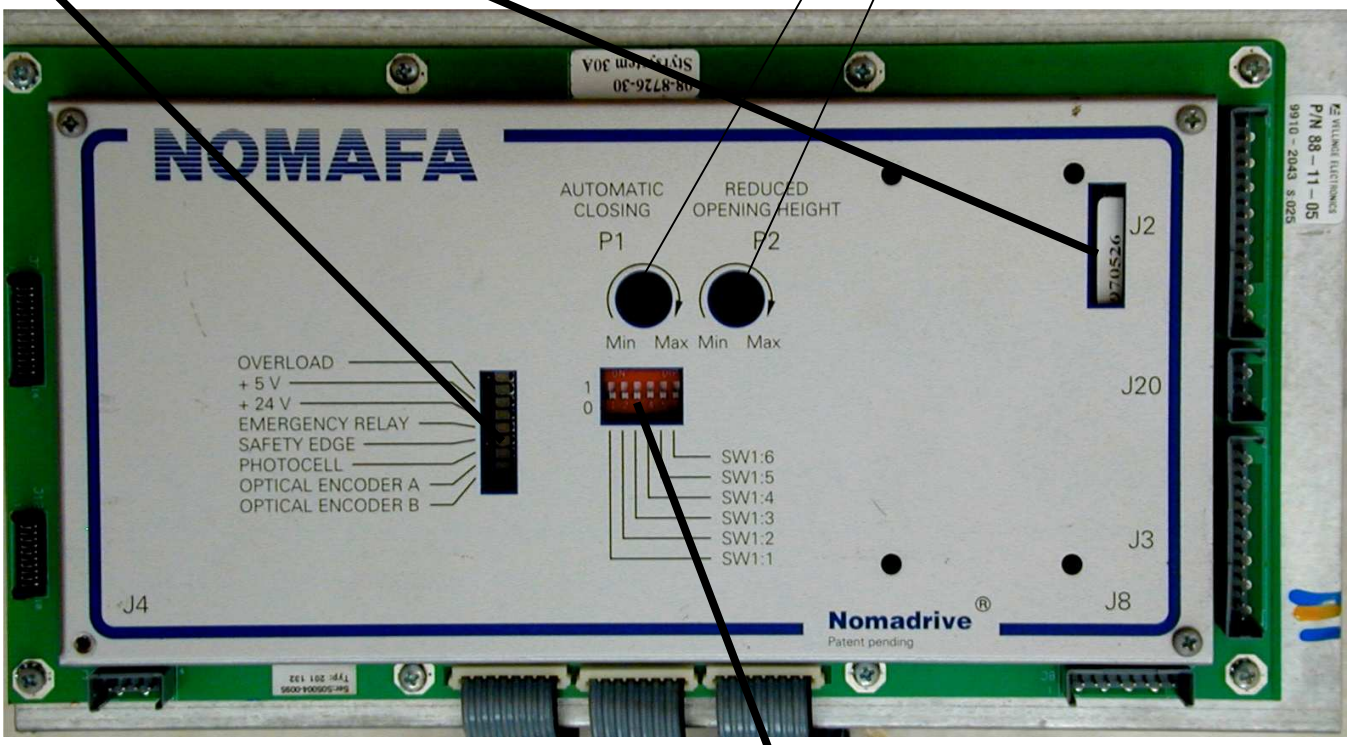
Where to find what on front plate of CPU:

LED indicators: 1 overload
2 5 volt supply
3 24 volt supply
4 emergency stop relay
5 safety edge
6 photocell
7 encoder channel A
8 encoder channel B

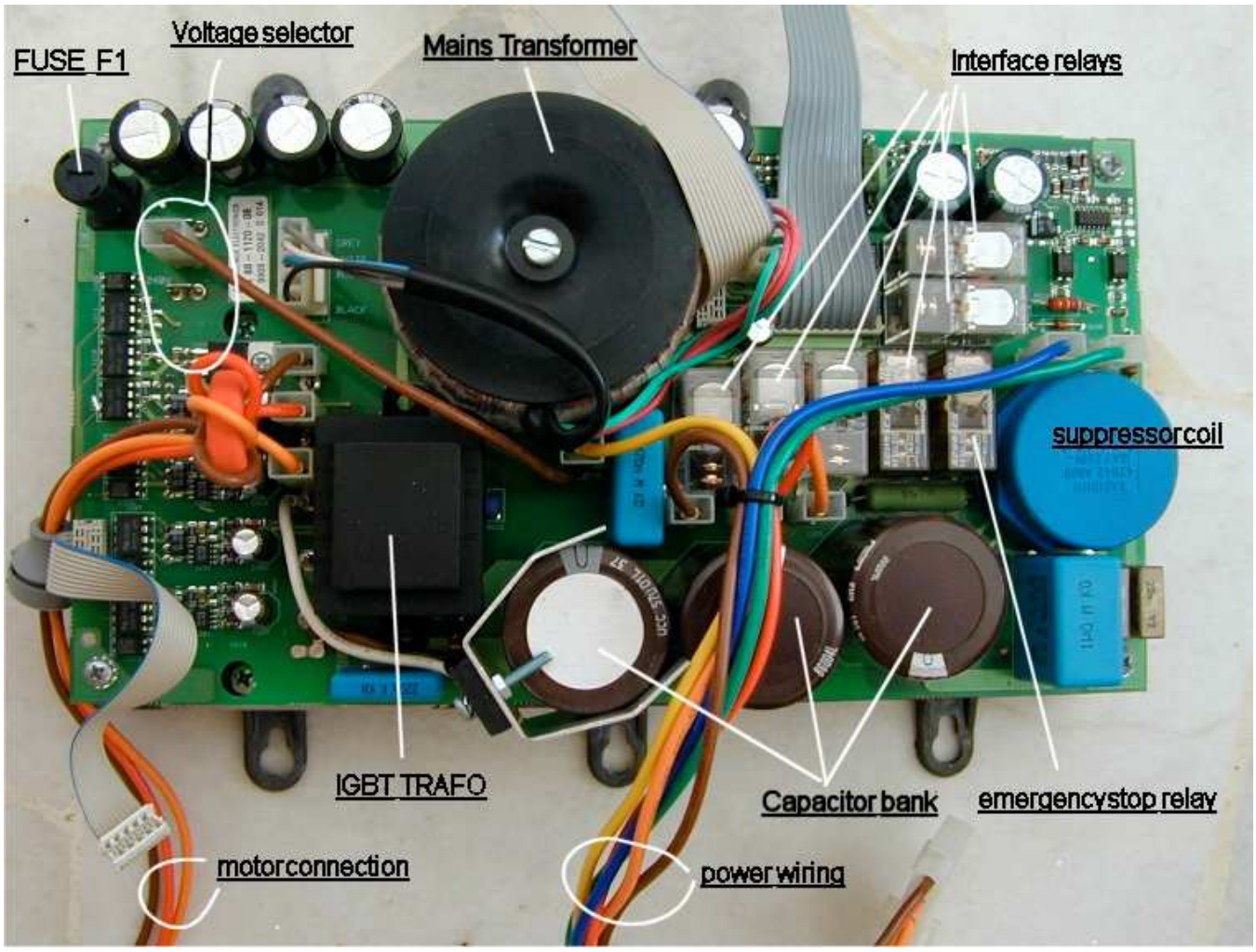
P1 = pause time delay 1-60 sec

P2 = Reduced openings height (see term. X1-31 & 32

EPROM window



1 Safety edge: Automatic closing after impact
2 Count down for closing after passing photocell (delayed closing)
3 Interlocking with other door. A must be close after B, also reversed
4 Photocell in Doorline
5 Traffic light flashing when door opens and closes + 2 s. before
6 Not used in standard program

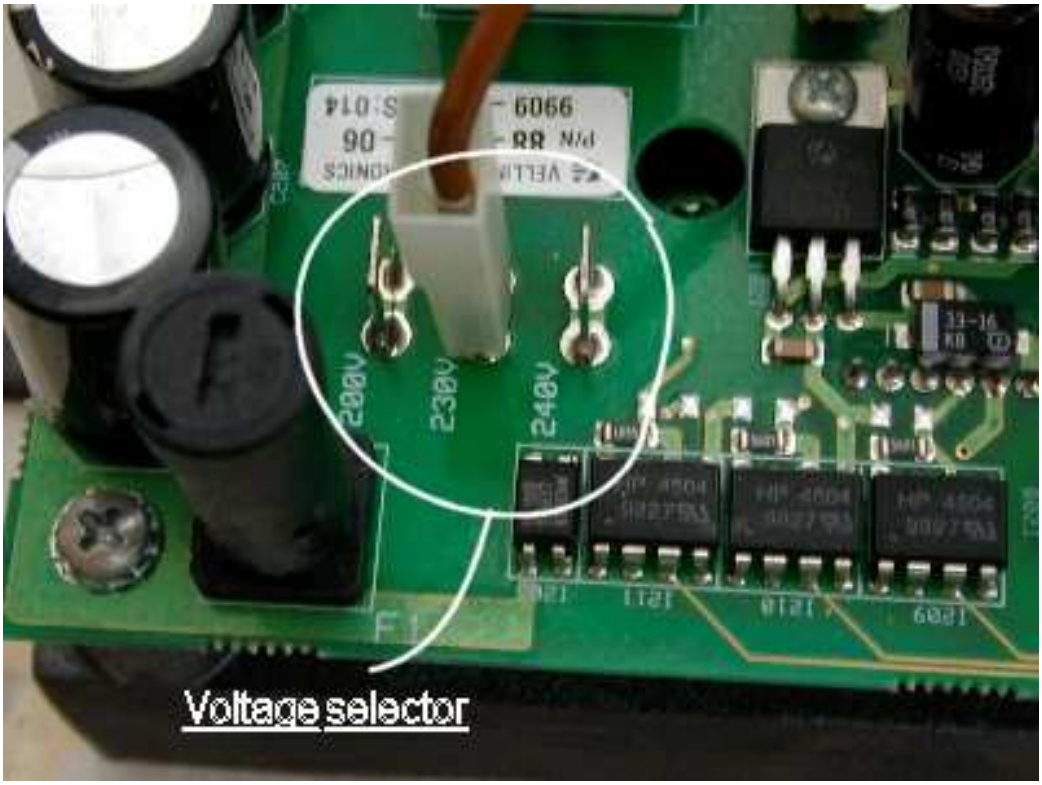


Motor card connections and lay-out (NomaDrive)

The motor card houses the following elements:

- IGBT module (between card and heatsink)
- Capacitor bank (storage of DC power)
- Interface and emergency relays
- Power supply trafo's for IGBT and Control Voltage
- Suppressor coil (anti-spikes)
- Voltage selector 200-240 VAC

WARNING: DO NOT REPAIR ON COMPONENT LEVEL !!!

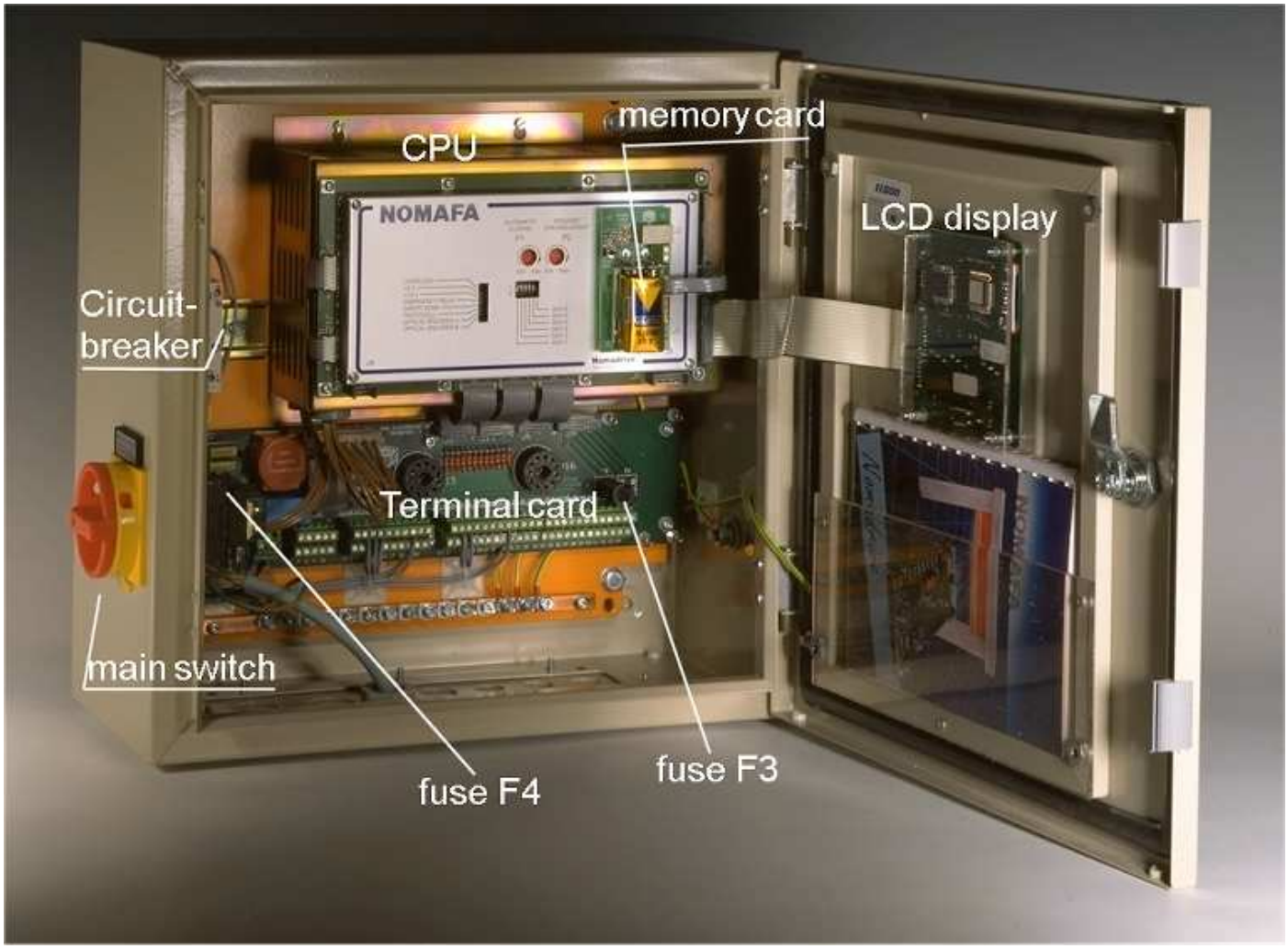


Voltage selector on motor card. (LH side of CPU)

(200 V – 230 V – 240 V AC) 50 or 60 Hz can be used.



EPROM (with program data)

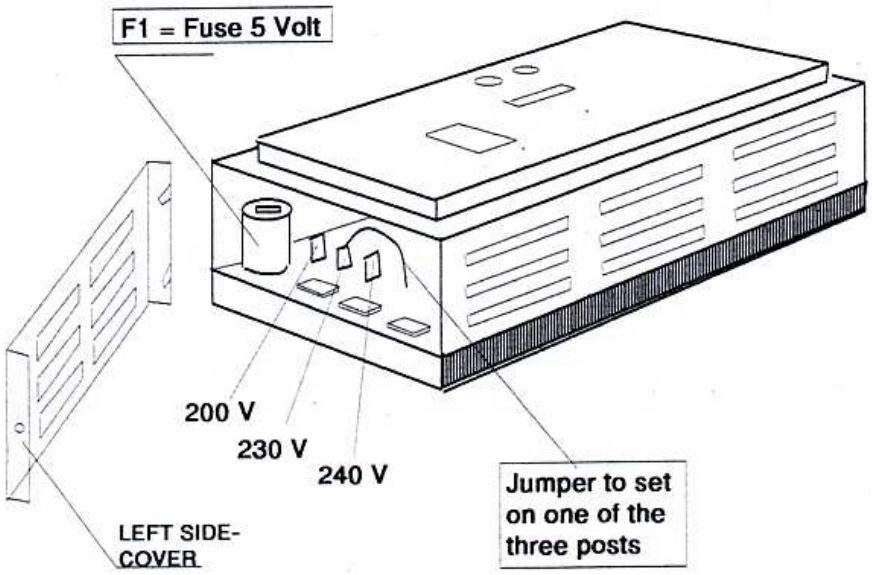


Control box interior:

- housing is IP 65 protection standard
- Cable glands are in the bottom plate of the housing
- Cabinet is mounted against the wall using spacers
- For the counter cabling, only use original screened cable
- For fuses use : 20 x 5 mm glass fuses standard type
- F4 = 2 A
- F3 = 1 A

Voltage selection on CPU

The below sketched Voltage selection with three (3) posts with the "jumper" cable and AMP connector, is meant to pre-set the CPU for different mains supply Voltages in different countries. All Voltages must be within 10% of the official Voltage rate. (IEE standard)



Correct setting :

1. **measure the mains supply Voltage with an accurate Voltmeter (digital) on terminal 4 & 5 of the terminal card.**

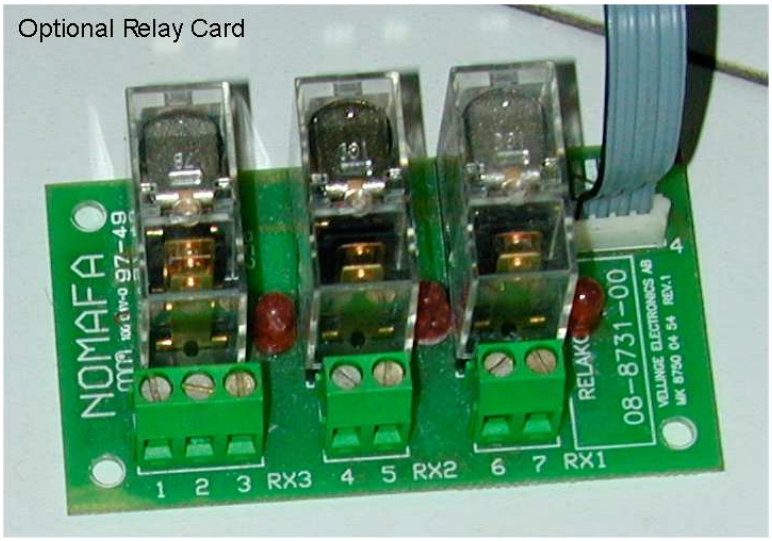
2. **set "jumper" cable on post closest to the measured Voltage of the supply Voltage.**
   (sample : measured Voltage --> 233 Volt , set jumper on 230 Volt)

3. **Start Nomadrive system with Power ON, look for error message 16, if no message appears, the setting is correct. If a message appears handle accordingly by setting higher or lower Voltages.**

ADJUST SETTING WITH "POWER OFF" AT ALL TIMES

PERIPHERALS :

Extra Relay Card:



Relay card terminals X3

RX1 = 6 & 7 (NO)

RX2 = 4 & 5 (NO)

RX3 = 1 (NO)
RX3 = 2 (NC)
RX3 = 3 (com)

Connector J4

This Relay card is used for:

- Extra relay contacts extension if so desired (status relay option)
- Airlock interface option
- Error message output relay option
- Interlock option with other non-NomaDrive doors
- Special program applications for automation purposes
- Etc.

Note: Features for the use of this card are programmable.
If special featured are required, please contact the engineering dept. or regional office of Albany Doors.

Pheripherals :

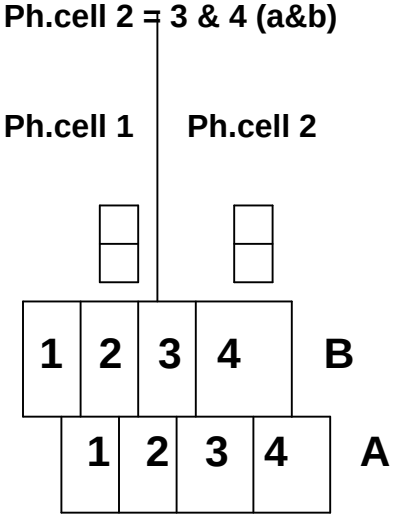


Photocell Card:

2 separate Photocell-amplifiers are on this card.

Adjust the pot.meter of each channel to the least needed sensitivity by turning the dial anti-clok-wise.

Ph.cell 1 = 1 & 2 (a&b)
Ph.cell 2 = 3 & 4 (a&b)

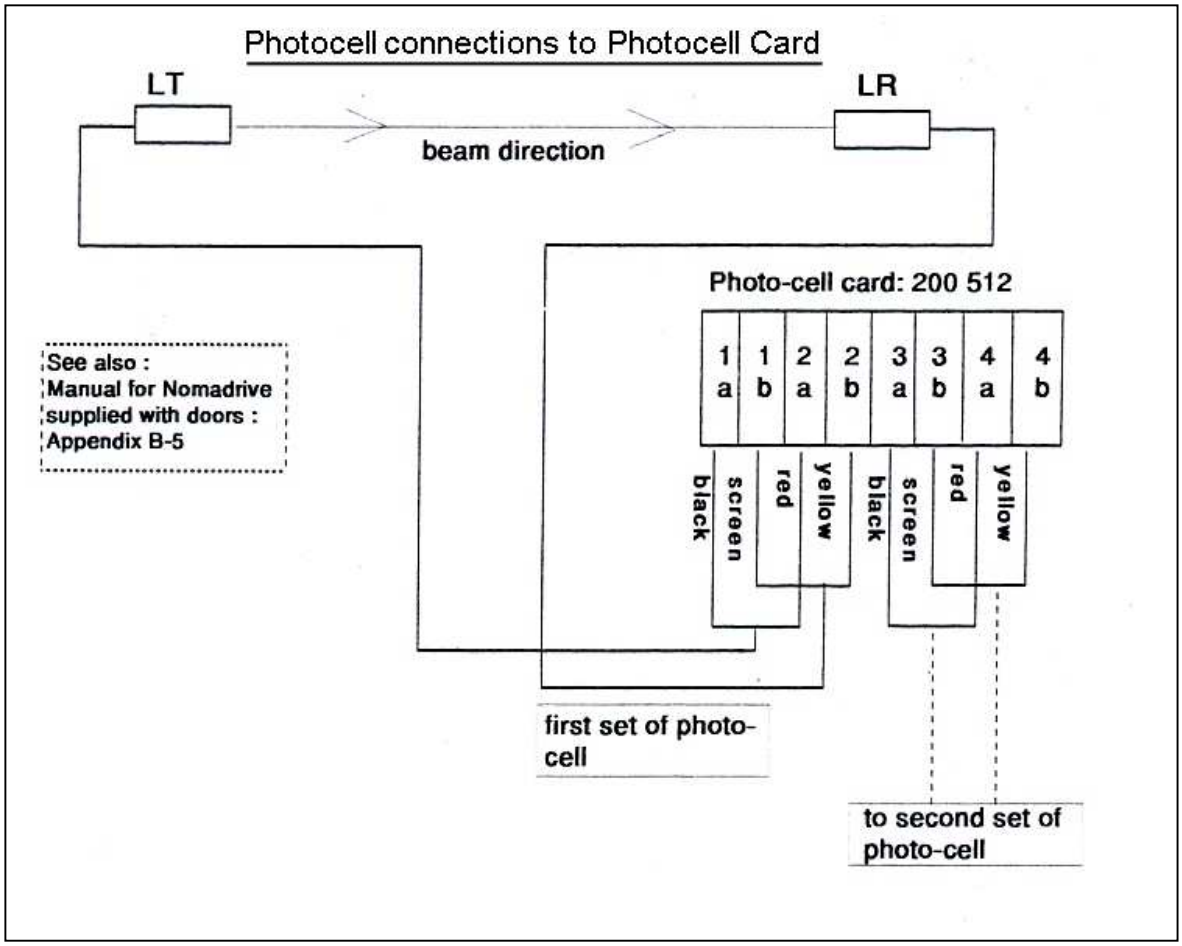


Be aware that the extra Photocell Card has to be programmed into the system by using the service instrument and installing the Photocell option in the Option menu.:

- Photocell in door line program Options: D – L
- Photocell NOT in door line. program Options: PH.C. 1 or 2

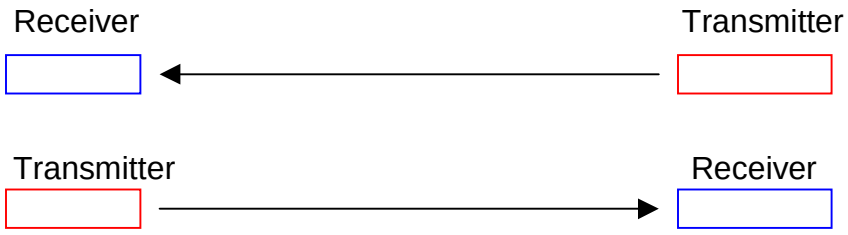
If several photocells are placed parallel, put transmitter and receiver of each set opposite to each other to prevent “cross-talk”

Peripherals: Photocells



Remark:

If two sets are used in parallel configuration, please use the sets with
The transmitters/receivers opposite to each other, to avoid cross interference.



Peripherals : J8 - Photocell

Connections on CPU for Photocell Card:

Flat ribbon connector J-8 :

1. **+ 24 VDC**
2. **+ 8 VDC**
3. **Logic output Photocell 01 (IPH1_X)**
4. **Logic output Photocell 02 (IPH2_X)**
5. **Enable / Disable Photocell 01 & 02 (IPHL_X)**
6. **N/A**
7. **+ 5 VDC**
8. **GND**

Note: It is important to look at the OPTION menu and check if the Status of the OPTION Photocell is programmed or not.

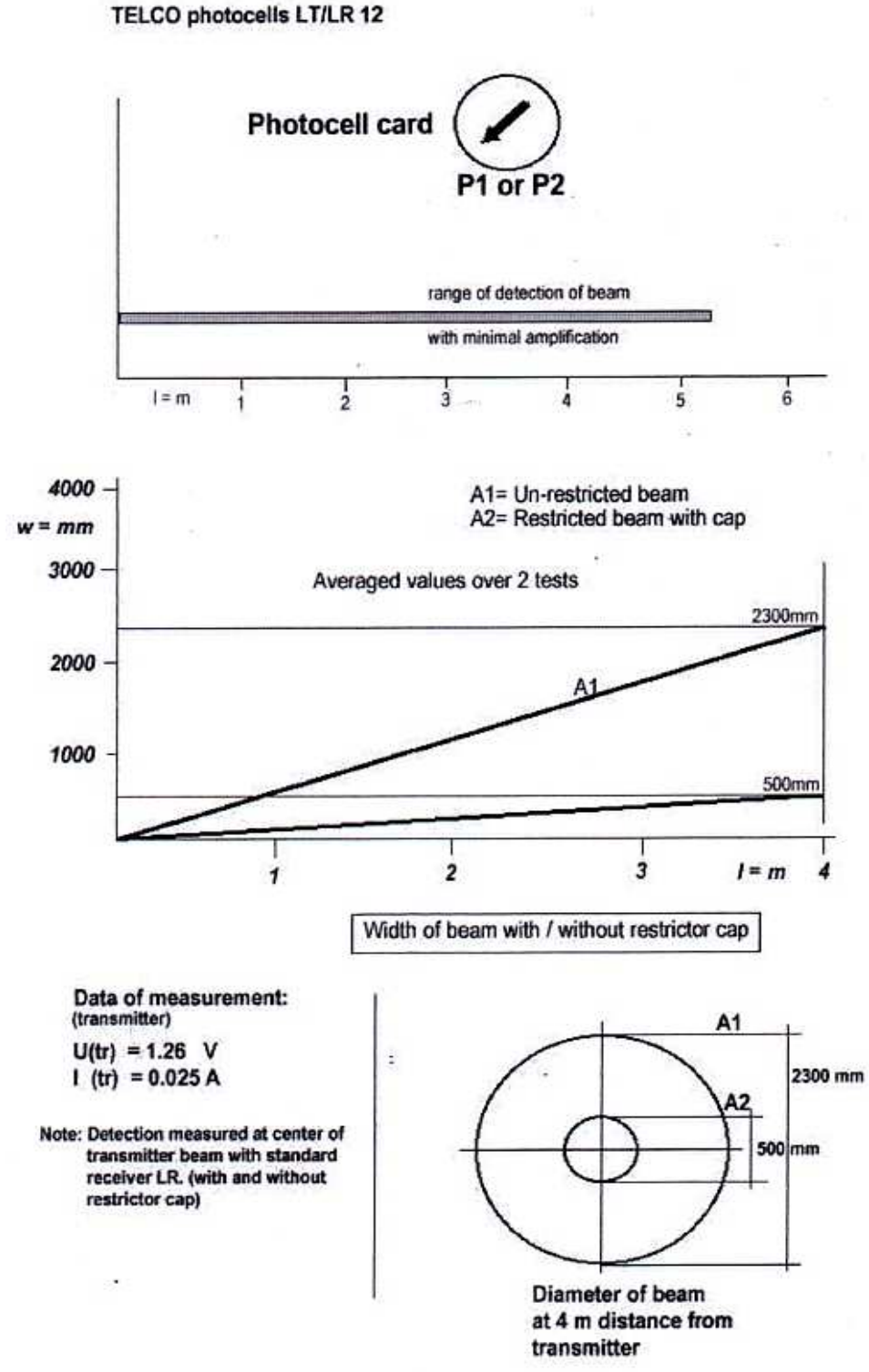
With a logic checker it is possible to diagnose the CPU In / Output for proper operation.

The Voltages across the J-8 connector are easily checked with a Digital Voltmeter.

WARNING: ONLY ELECTRONICALLY QUALIFIED TECHNICIANS SHOULD CARRY OUT SUCH TESTS !!!

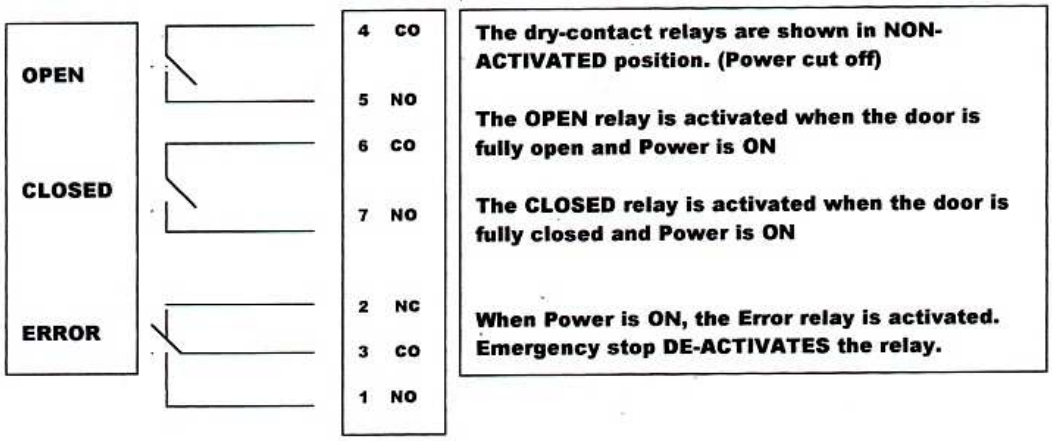
See also page #20 in the book : Tips and tricks for Nomadrive

Peripherals: Telco Photocells



Peripherals: Status relays

Use of STATUS relays in doors with *Nomadrive* :
(on Relay card X3)



The dry-contact relays are shown in NON-ACTIVATED position. (Power cut off)

The OPEN relay is activated when the door is fully open and Power is ON

The CLOSED relay is activated when the door is fully closed and Power is ON

When Power is ON, the Error relay is activated. Emergency stop DE-ACTIVATES the relay.

Note: 24 VDC IS CUT-OFF from all inputs when the Emergency, Hand-crank or Thermo-switch is activated.
- **BUT NOT THE STATUS RELAY POWER.**
- **BOTH STATUS RELAYS WILL STILL BE SHOWING THE REAL STATUS OF THE DOOR.**
- **The ERROR summary relay is de-activated**

To monitor in a reliable way the door with an external PLC for conveyor or airlock use, always use the OPEN/CLOSE relays signals together with the ERROR summary relay.

ALL status relays are giving a constant signal during the time that a condition exists.

A STOP signal via 23, 24 does not cause an error and will NOT de-activate the Error-summary relay.

The standard STATUS relays : 42 OPEN, 43 CLOSED on the terminal card X1 are acting exactly the same way as the STATUS relays on X3 relay card.

Configuration

Terminal	Name	Description
1 2 3	Motor	200, 230, 240 VAC
4 5	Supply from the net or from the transformer.	200, 230, 240 VAC 50/60 Hz.
6a 6b	Traffic light	220 VAC. Switched on when the door is closed, in motion and 2 sec before time-delayed closing. See also SW 1.5.
7a 7b	Traffic light red	220 VAC. Switched on when the door is closed, in motion and 2 sec before time-delayed closing. See also SW 1.5.
8a 8b	Traffic light green	200, 230, 240 VAC. Switched on when the door is open.
9a 9b	190 VDC	Supply to the fixing brake, rectifier.
10a 10b	200, 230, 240 VAC	Supply to the activators etc.
11a	Optical encoder	Channel A
11b	Optical encoder	24 VDC +
12 a	Optical encoder	Channel B
12 b	Optical encoder	24 VDC -
13a 13b	Emergency stop 1	Cuts off the supply to the static frequency changer. Reset by a new open-impulse.
14a 14b	Emergency stop 2	Cuts off the supply to the frequency converter. Reset by a new open-impulse.
15a 15b	Crank switch	For crank on motor. Bridge if the crank switch is not connected.
16a 16b	Thermostat	Built-in thermostat in motor winding. Disconnects at tedious overloading of the motor.

Terminal	Name	Description
17a 17b	Safety edge Pneumatic safety edge	If the safety edge is affected when the door is closing, it reverses to open position and you have to give new close impulse. See also SW 1.1. For each door cycle the safety edge is tested when it touches the floor. If the safety edge does not work the door goes over to hold to run mode the next time it opens. Then you must close the door to receive test impulse at floor. Test impulse also sets the optical encoder to 0.
18a 18b	Position 50 mm	Extra limit position for disconnection of the reversing function of the safety edge at pneumatic and optic safety edge (not touchless). Must not be adjusted higher than 50 mm above the floor.
19a 19b	24 VAC	Supply to activators etc.
20a 20b	24 VDC - 24 VDC +	Supply to activators etc.
21a 21b	Open	Open impulse. Re-sets after error.
22a 22b	Open	Open impulse. Re-sets after error.
23a 23b	Stop	Stop-impulse. Bridge if stop buttons are not connected.
24a 24b	Stop	Stop-impulse. Bridge if stop buttons are not connected.
25a 25b	Close	Close-impulse. Re-setting after error if the door is open.
26a 26b	Close	Close-impulse. Re-setting after error if the door is open.
27a 27b	Open / close	One impulse for opening, one for closing. Re-sets after error.
28a 28b	Open / close	

Terminal	Name	Description
29a 29b	Opening with time-delayed closing.	At an impulse the door opens and then closes after set time on potentiometer P1, 0 - 60 sec. The count-down starts when the door is totally open. The time gets updated if a new impulse is given. Re-setting after error.
30a 30b	Opening with time-delayed closing.	At an impulse the door opens and then closes after set time on potentiometer P1, 0 - 60 sec. The count-down starts when the door is totally open. The time gets updated if a new impulse is given. Re-setting after error.
31a 31b	Opening to reduced height with time-delayed closing.	Same function as terminal 29-30 but opens to the height you have set with potentiometer P 2.
32a 32b	Opening to reduced height with time-delayed closing.	Same function as terminal 29-30 but opens to the height you have set with potentiometer P 2.
33a 33b	Safety radar or safety photocell	Reverses the door, which closes again after delay pre-set at P1.
34a 34b	Extra in-put	Not used in the standard system.
35a 35b	Brake away switch	Stops door when bottom beam brakes away after hit
36a	Photocell in doorline, Transmitter (black)	If the beam of the photocell disconnects when the door closes, the door reverses and closes after set time on P 1. Se also SW 1.2 and 1.4.
36b	Photocell in doorline, Receiver (screen)	
37a	Photocell in doorline, Transmitter (red)	
37b	Photocell in doorline, Receiver (yellow)	
38a 38b	Inductive loop 1	Same function as terminal 29 - 30 or terminal 36 - 37. Is selected by the service instrument.

ATTENTION: In case of Automatic Dis-engagement Devices Applied to ACS70 control systems : Check page 38 for changed Terminal card connections of the brake coil of the drive motor !!!

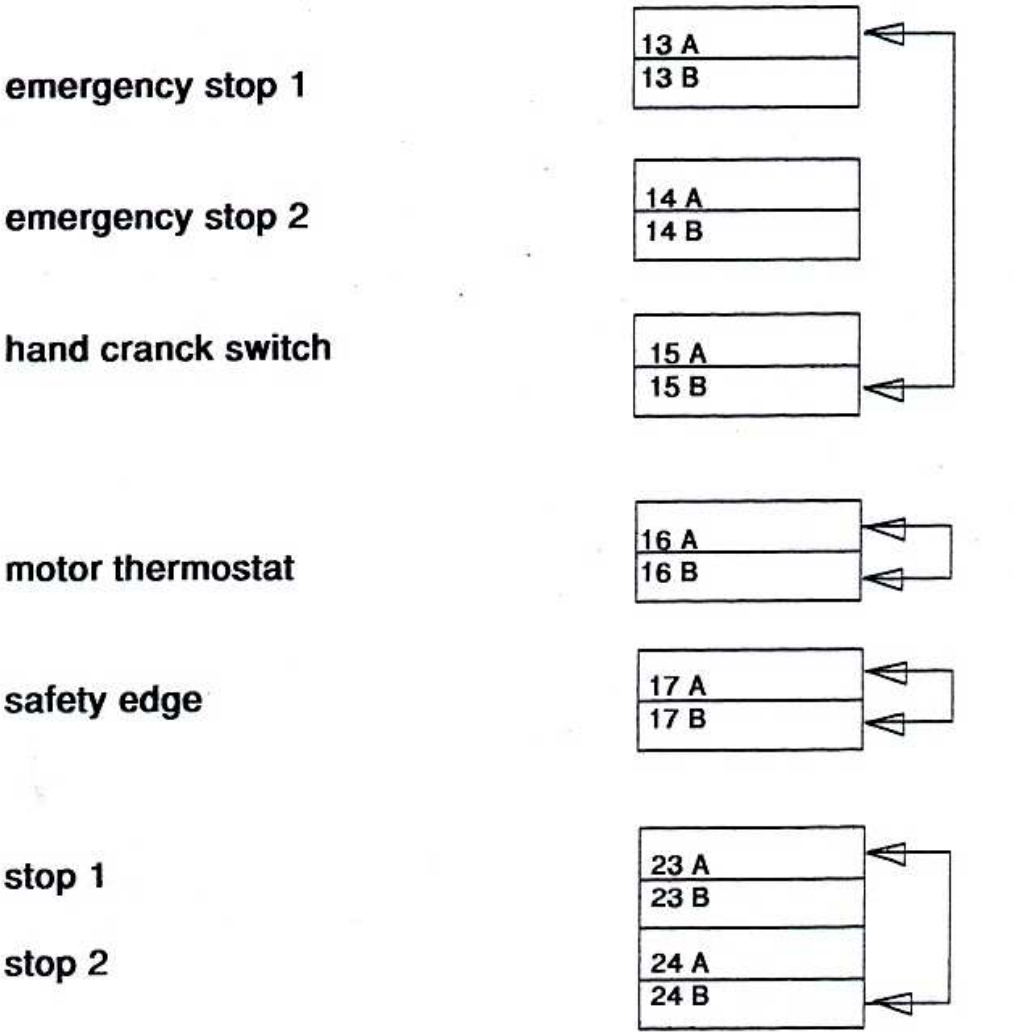
Terminal	Name	Description
39a 39b	Inductive loop 2	Same function as terminal 29 - 30 or terminal 36 - 37. Is selected by the service instrument.
40a 40b	Analogue in-put 1 Analogue earth	0 - 10 VDC 10 kOhm. Not used in the standard system.
41a 41b	Analogue in-put 2 Analogue earth	0 - 10 VDC 10 kOhm. Not used in the standard system.
42a 42b	Door open	Relay contact which closes when door is open. Max. 24 V.
43a 43b	Door closed	Relay contact which closes when door is closed. Max. 24 V.
44a 44b	Door B closed	Relay contact from interlocked door which closes when the door is closed.

Indications on the circuit card

Overload	Indicates that the frequency changer has been released for overloading, undervoltage or over temperature.
+ 5 V	Indicates supply to the microprocessor.
+ 24 V	Indicates supply to the inputs.
Emergency relay	Indicates that the emergency stop relay is activated.
Safety edge	Indicates that the safety edge is unaffected.
Photocell	Indicates that the photocell in the door-line is unaffected.
Optical encoder A	Indicates channel 1 on the optical encoder.
Optical encoder B	Indicates channel 2 on the optical encoder.

SEE ALSO NOTE ON PAGE 38

NECESSARY CONNECTIONS : *NOMADRIVE*

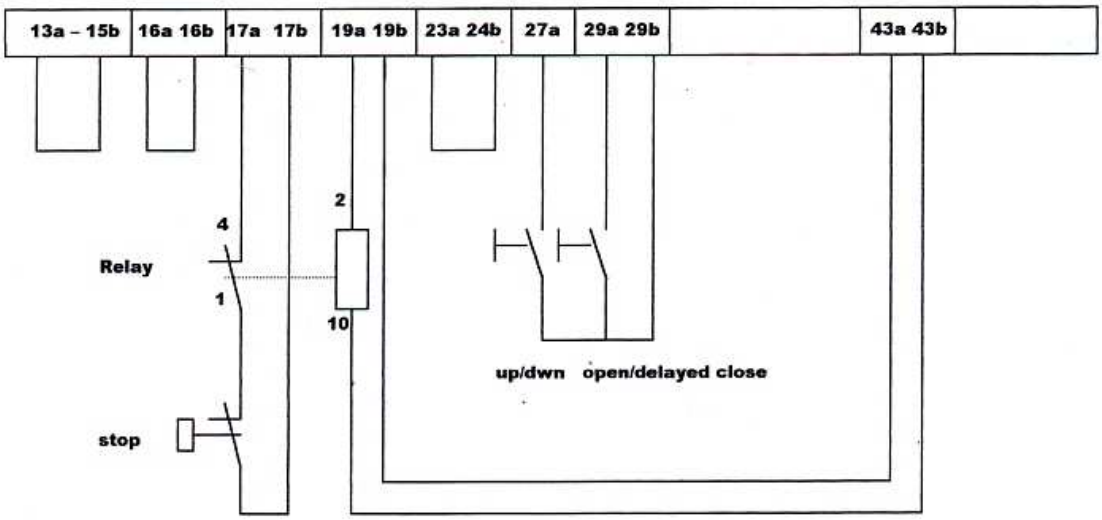


All these contacts can be "bridged" by a "jumper"

A "jumper" is a piece of wire, to connect two terminals

without these contacts "CLOSED" :
NOMADRIVE will NOT operate !!

NOMADRIVE SET-UP FOR CPU-TESTING:



The above configuration uses :

- **1 pcs. 11 pin relays 24 vac.**
- **1 pcs. Push-button box with 2 no pushbuttons (up/dwn -- open delayed close)**
- **1 nc pushbutton (stop)**

Furthermore:

The motor + decoder should be connected to the terminal card.

The relays "dummies" the safety edge when going down, by using 43a+b as impulse at lower limit position.

The Stop push-button is for re-set purposes and testing of the safety-edge position. (specially needed with non-standard EPROM's.)

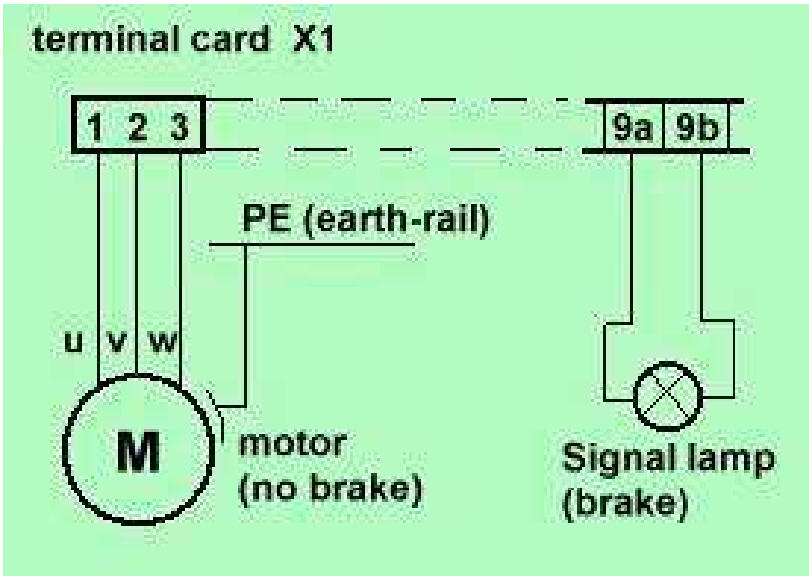
The CPU should run without any dip-switch set, or any option invoked, unless test of such option is required.

The motor may be substituted by a small 0.4 HP motor (3 phase mimicked by using a capacitor between phase 2 & 3), thereby enabling to run the set entirely on 220-240 single phase supply.
The decoder disc should be fitted directly onto the 10 mm shaft of such small Ac motor. (the disc is standard 10 mm diam.)

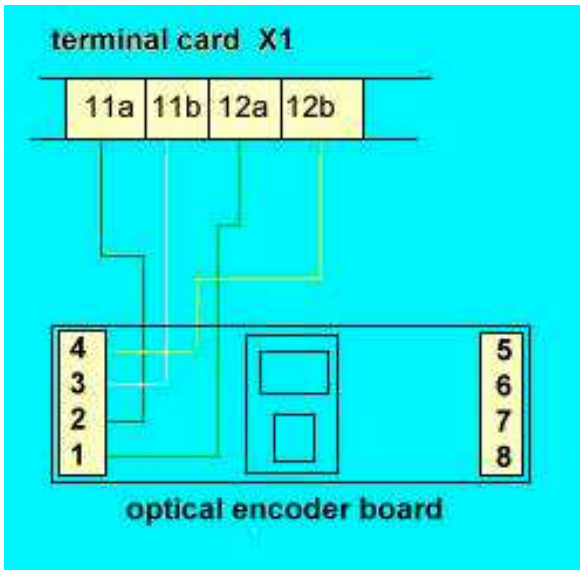
The whole contraption can be placed on a plate of multiplex and fitted with a mains supply cable of sufficient length. (motor + terminal card + pushbuttons + relay.)

It makes up for a very portable test unit, also suitable for educational purposes.

The following connections are necessary for the test set-up with a small Electric motor. The same applies when a NomaDrive unit is tested.



The signal lamp replaces the brake during the test when a NON NomaDrive motor is used.

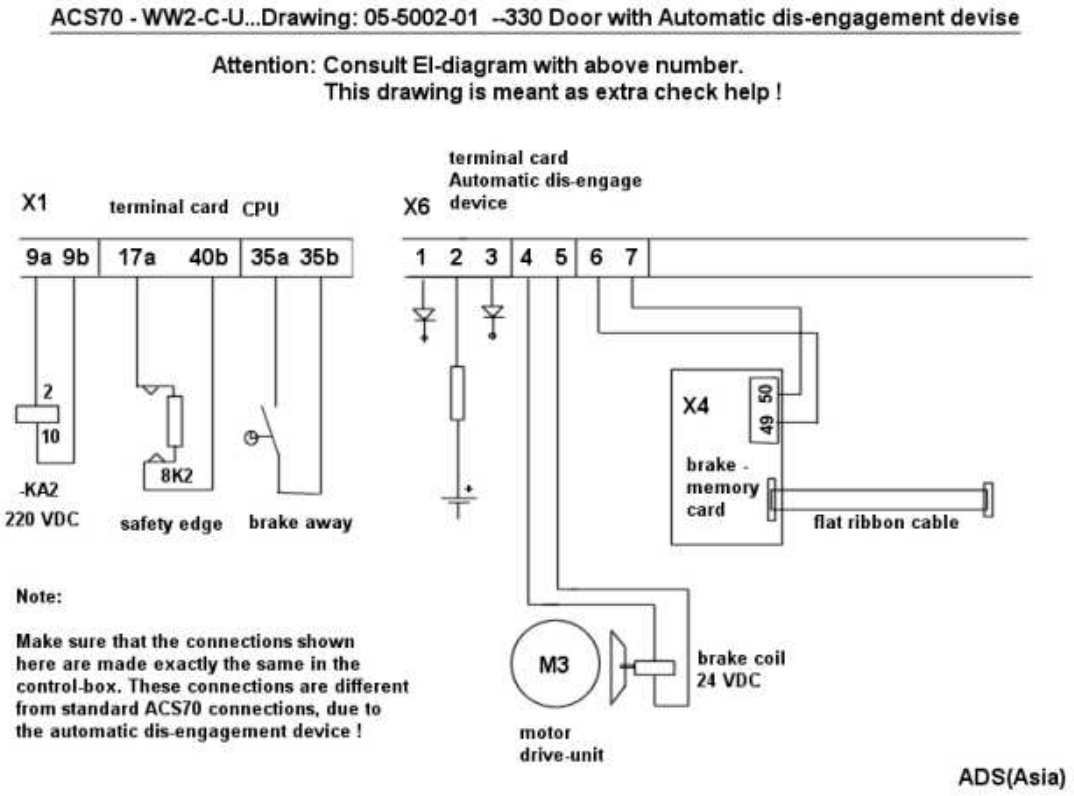


Encoder connections to the terminal card

The encoder disc has to be put (with standard key) on the 10 mm thick shaft of the small motor. (with an original NomaDrive motor it is integrated)

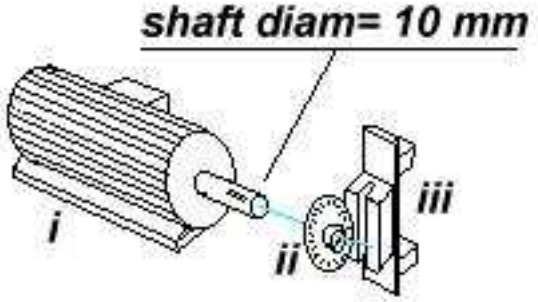
ATTENTION : BE CAREFUL → DIFFERENT TERMINAL CONNECTIONS

In case of Automatic dis-engagement devices on 330 model doors, the terminal connections from the standard ACS70 terminal card are re-routed via the Battery Back-up system in the control-box, according to the below diagram !!!



Note:

The brake coil connections X1-9a+b are NOT connected to the motor brake coil, but connected to relay KA2. The Motor brake coil is directly connected to X6-4+5 from the dis-engagement device. Make sure you consult the Electrical diagram shipped together with the original door, and consult also the installation manual of the door which has to be tested.



i. Motor (0.4 kW-220 VAC-3-ph)
ii Encoder disk (03-0050-25)
iii Pulse Transducer board (03-0051-25)

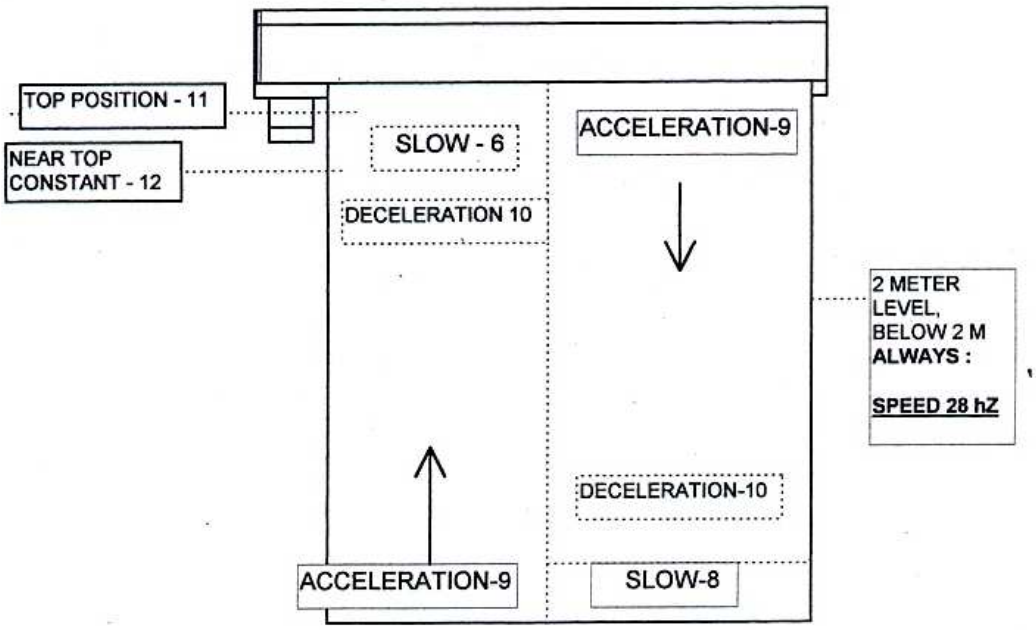
Note: Fit the encoder disc and optical encoder board within a plexi-glass cover to Avoid un-safe working conditions with your hands.

The test unit should be used with the following equipment:

- LCD Display hooked onto J-3
- Terminal card hooked up to : J1, J11, J6, J9, J10, J18
- Motor on J 21 (U. V. W.)
- Power 200-240 VAC from mains on J22 (N + L1)
- Good PE connection on CPU and mains cord
- Service instrument on J2 (if connected, LCD display is switched off)
- Use an external circuit breaker for safety between mains and terminal card

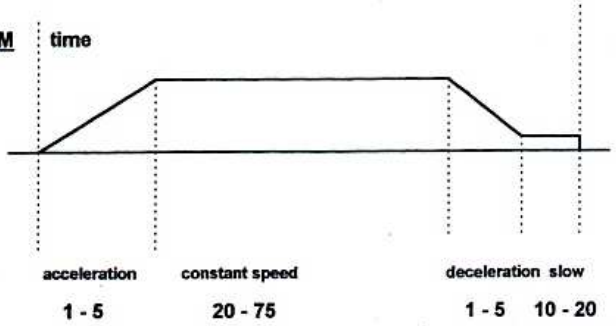
Materials:
- Motor 200-240 VAC-3-ph (0,2 ~ 0,4 hp is enough)
- Terminal card: 08-8717-14 (or higher)
- Encoder board 03-0051-25
- Hub of disk 03-0052-25
- Encoder disc 03-0050-25
- CPU (to be tested)
- Signal lamp 220 V-15 Watt (incl. Socket)
- 24 VDC relay 11 pens (including socket , locally available)
- Insulated plate 50 x 30 x 1,5 (w x l x d)cm
- Push button set: up/down in housing , locally available)
- PVC wire 1,5 mm2 multi-core, for wiring of board

PROGRAMMING POSITIONS FOR "Noma*Drive*"



SETPOINT :

OPENING DIAGRAM



- **5. Set speed Up**
- **6. Set slow speed Up**
- **7. Set speed Down**
- **8. Set slow speed Down**
- **9. Set acceleration time**
- **10. Set deceleration time**
- **11. Set top position distance**
- **12. Set nearly top constant** (A constant which conditions the way the door stops at the top

For all values in SETPOINT, refer to the parameter table provided by Nomafa AB



Service Instrument is used to program the parameters of each door individually.

The Service instrument has a cable to be connected to J-2. The connector at the right-hand upper corner of the CPU.

As soon as the Service Instrument is connected to the CPU, the existing LCD display on the control box is switched off.

With the Service Instrument, the following functions can be used:

1. STATUS (The status of all internal connections can be monitored)
2. SETPOINT (the parameters of the door program can be adjusted)
3. OPTIONS (optional auxiliaries / activators can be installed)
4. INSTALL (a program can be INSTALLED or DELETED)



The connector for the cable of the Service instrument on the CPU

The Door has the following default parameter setting:

- LANGUAGE GB
- COUNTRY OTHER
- TYPE 200/300 OTHER
- MOTOR 1.1 KW
- SET SPEED UP 50 HZ
- SET SLOW SPEED UP 10 HZ
- SET SPEED DOWN 50 HZ
- SET SLOW SPEED DOWN 10 HZ
- SET ACCELERATION TIME 3
- SET DECELERATION TIME 3
- SET TOP POSITION DISTANCE 1500
- SET NEARLY TOP CONSTANT 7
- SET NEARLY BOTTOM POSITION 100
- COUNTER 0000

PROGRAMMING CPU

PROGRAMMING NOMADRIVE CPU ----> SERVICE INSTRUMENT

The "Service Instrument" enables the Nomafa technician to "tailor" the Nomadrive system to "behave" according to the demands of the Door-type, Application , Activators & Options.

The Service Instrument can be hooked-up to the CPU with the help of its cable and connector, to be inserted into the 16-pins connector at the upper right hand corner of the CPU.

The Service Instrument has a "Display" on which a "Menu" is shown. This "Menu" tells the user which "Function" or "Data" or "Group of Functions" can be programmed or looked at.

The Service Instrument has also 6 (six) "Keys" :

1. - **(TEXT)** key : Moves from one "Text-step" to the next "Text-step"
2. - **(+)** key : Moves inside a "Menu" from left to right and also "Increases" values with 1 (one)
3. - **(-)** key : "Decreases" values with 1 (one)
4. - **(SAVE)** key : Stores updated values into the EEPROM (old values are overwritten)
5. - **(MENU)** key : If pressed, it returns to the "Main-Menu" regardless in whatever Text-step one is, also shifts between "menu" & "alarm" texts
6. - **(ERASE)**key : can erase a complete "Installation-setup"

DO NOT USE "ERASE", UNLESS SERIOUSLY NEEDED

EVERY NEW DOOR HAS A "DEFAULT" PROGRAM ALREADY INSTALLED

SERVICE INSTRUMENT (continued)

There are 4 Menu's :

1. - **STATUS** : To check the status of :
   - Activators
   - Door position
   - Options
   - Outputs

2. - **SETPOINT** : - Set program parameters for :
   - Language , country , type of door
   - type of motor
   - speed , acceleration , deceleration
   - position of door
   - nearly top constant

Every single step of this menu enables the user to adjust the parameters for the drive-unit

- type of motor
- speed , acceleration , deceleration
- top- & bottom-position & nearly top constant

3. - **OPTIONS** : To select options and activate their functions

4. - **INSTALL** : - To check if the door has been provided with Program data that has been initiated.
   - To erase existing "setup" data from the EEPROM (Be very careful with using this possibility)

Note : In practice the programming of the Nomadrive unit is mainly limited to :
- set top position adjustment (point 11)
- set acceleration & deceleration (point 9 & 10)
- set speed up (point 6)

Program first of all "top position" (11) and run the door many times, observing all mechanical behaviour of the door. Only after being satisfied that the door is running correctly program the other parameters, using the latest service-information sheets available on this subject from NOMAFA AB.

DEFAULT PROGRAM PARAMETERS

An already mechanically installed door, and with all the external wiring connected to the NOMADRIVE unit, must be checked as follows:

1. - Check if the "Safety-Edge" is working by checking if LED on the CPU turns ON & OFF
2. - Check with Service instrument in **INSTALL** mode, if program has already been installed from the Factory. If so, proceed with **INSTALL** procedure. If **NOT**, continue the following list.
3. - Put Nomadrive system into "Hold to Run" mode and CLOSE the door till the Safety-edge hits the floor.
   (HOLD to RUN mode should terminate)
4. - Switch Control box Display into "TEXT" mode and proceed to text step 2 : Position of Door in Pulses
5. - Put Nomadrive system into "Hold to Run" mode again, and run the door UP all the way, note the position of the door in Pulses on piece of paper, and run the door DOWN again.
6. - After the Safety-edge hits the floor, the Hold to Run mode terminates.
7. - Connect the Service instrument to the CPU.
   (Now text Display on the Control box is blanc)
8. - Choose Menu "**SETPOINT**" and proceed to the following steps:

The below mentioned “Menu” for the Service Instrument enables the user to dig deeply into the operational details of the CPU and control system.

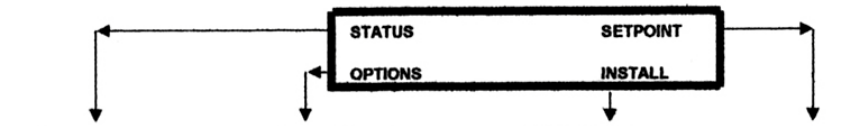
The STATUS function will help the fault-finding to be easier. Each and every single input can be tested by looking at the requested input number and using the activator or sensor on that input to change its status.

The OPTION function enables the user to install or de-install all the optional available activators and safety devices in the system.

Special attention for the selection 5 : INPUT 35 NC
This safety edge input for electrically controlled contact edges can be switched from NC to NO by marking or un-marking the square on the screen.

SETPOINT enables the user to fine tune the default parameters of the door, given from the factory. Also the Language is adjustable to 3 other languages.

Menu structure for the Service Instrument



STATUS

9 STEPS
each terminal
connection
can be tested

1 EMERGENCY 13-15
TEMP 16
STOP 23-24
SAFE 17
POWER EM.REL

2 OPEN 21-22
CLOSE 25-26
OPEN/CLOSE 27-28
AUTO 29-30
AUTO R.H. 31-32

3 PH.C.S. 36-37
LOOP 1 38
LOOP 2 39
50 MM SW 18

4 PH C. 1. 1-2
PH C. 2. 3-4
SAFETY PH. C. /
RADAR 1-2#

5 EXTRA INPUT
#1 34
#2 35

6 OPEN 42
CLOSED 43
INTERLOCK 44

7 BRAKE 9
LIGHT RED 6-7
LIGHT GREEN 8

8 RX1 6 - 7 RELAY CARD
RX2 4 - 5 RELAY CARD
RX3 1 - 3 RELAY CARD

9 OPENING &
CLOSING TIME
(IN SEC. ##)

OPTIONS

5 STEPS
to initiate

1 PH.C.1
FUNCTION
OPEN
D - L
SE
STS

2 PH.C.2
FUNCTION
OPEN
D - L
SE
STS

3 LOOP FUNCTION
OPEN
SAFE
DIR

4 ELECTRICAL SAFETY EDGE
NO MODULE

5 INPUT 33 NC

INSTALL

1 STEP

1 ALLREADY INSTALLE

RESET INSTALLATION

setpoint

14 STEPS

1 LANGUAGE
S / GB / D / F

2 COUNTRY
AT / CH
OTHER

3 200 / 300 / RAPIDT
OTHER

4 MOTOR
0.55 / 0.75 / 1.1 / 1.5 KW

5 SET SPEED UP ## HZ

6 SET SLOW SPEED UP ## HZ

7 SET SPEED DOWN ## HZ

8 SET SLOW SPEED DOWN ## HZ

9 SET ACCELERATION TIME #

10 SET DECELERATION TIME #

11 SET TOP POSITION DISTANCE ####

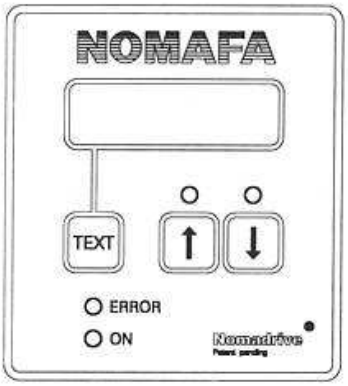
12 SET NEARLY TOP CONSTANT ##

13 SET NEARLY BOTTOM POSITION ##

14 COUNTER XXXX

Panel display

The panel display is placed in the front of the control box and has a display which is switched off in normal operating position. If an error should occur, the "ERROR"-indication is flashing and an error message is displayed. See Error codes chapter 5. If more than one error should occur the errors are displayed in priority order with the most serious error first. When this error has been attended to the next error is displayed and so on until all errors have been attended to.
Certain types of error cause control to go over in manual mode. Indicator lights above the control buttons will light, showing that the buttons should be used to open and close the door. Once the error is remedied the buttons can no longer be used, and the lights will go out.



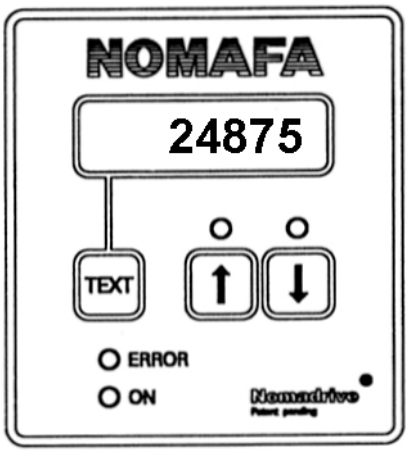
To place the door in manual-control mode, hold both buttons down for 2 seconds. Control returns to normal mode when the door has closed and its safety edge touches the floor. This is useful if the door's position is incorrect for some reason, such as a change of drive motor or if the door blade has been moved during a power cut. Manual mode does not work when emergency stop or cut-out are engaged.

If you press the [TEXT] button for 2 seconds the display is switched on and displays the number of door cycles. By pressing the [TEXT] button repeatedly the following is displayed:

1st	time	The position of the door
2nd	time	Time for automatic closing
3rd	time	Height for reduced opening height in no. of pulses
4th	time	Opening and closing time
5th	time	Frequency
6th	time	**Set value at upper limit. If you need to adjust the value, press [↑] and [↓] simultaneously for 10 seconds, until the 1st point in the display lights up. Press [↑], and the 2nd point will light up. Press [↓] and the 3rd point lights. Now the value can be increased by pressing [↑], or decreased by pressing [↓]. When the desired value is displayed, press [TEXT] to store the value and return to the first display.**
7th	time	Back to the number of door cycles again, etc..

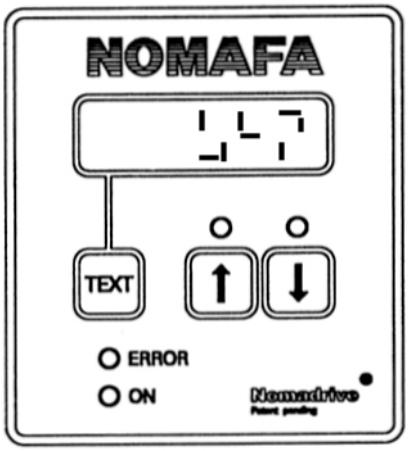
The display switches off again if the [TEXT] button is pressed for 2 sec. (The display should normaly be switched off.)
The LED "ON" is switched on when the control is in operating position.

Choice of display type
Press the [TEXT] button for 10 seconds, and the indicators will light up. Hold the [TEXT] button down while pressing the [↑] button for <klartext??> display, or the [↓] button for LCD display. Contrast can be adjusted with the potentiometer on the back of the control box.



Normal text picture when the Door is used with an LCD display

In this mode the Display shows numerical mode.



This Display shows the wrong setting of the LCD Display.

Press TEXT button for > 10 sec. And the indicators will light up.

Hold TEXT button DOWN while pressing the ↑ button for text mode , or the ↓ button for numerical mode.

The above mentioned modes are sometimes changed by the people working with the doors. Un-willingly some people change the display modes by just toggling the buttons at random and thereby causing this error.

Hardly ever will the display brake down. Mostly these error images are human behavior errors.

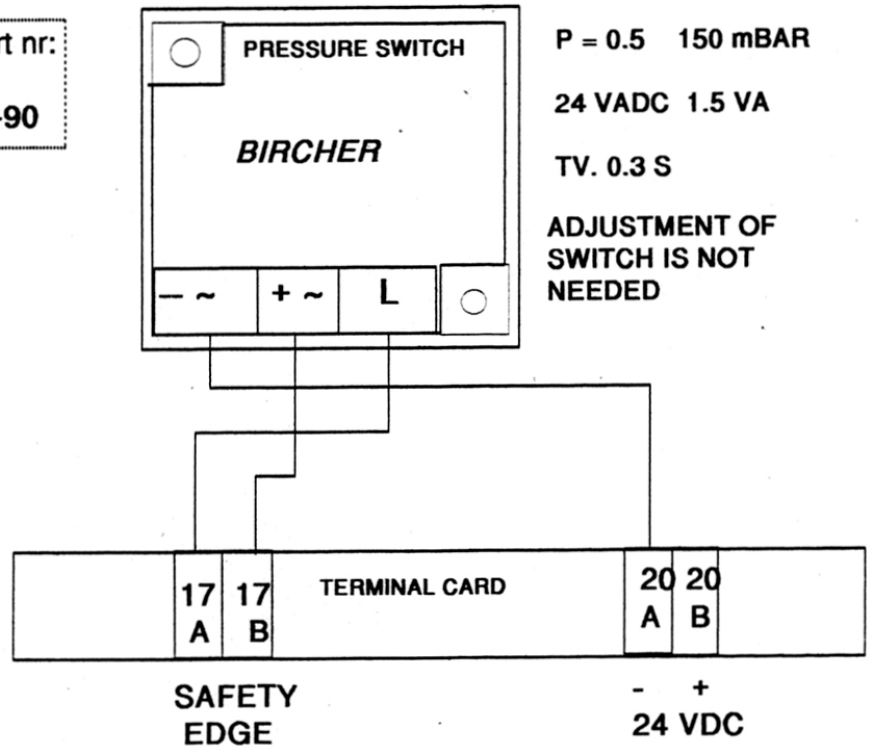
BIRCHER ELECTRONIC PRESSOSTAT

TECHNICAL INFORMATION : SAFETY EDGE D2-11 KV BIRCHER

The electronic safety edge switch (pressure switch) has to be connected to the Nomadrive terminal card (X1) as follows :

Spare part nr:

96-6015-90



WARNING :

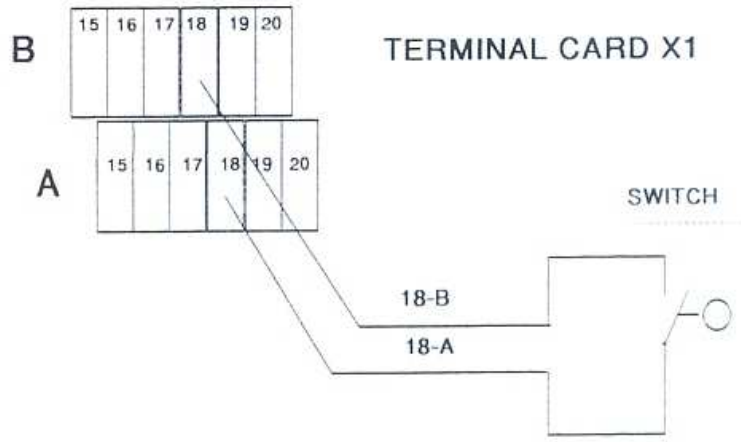
WRONGLY CONNECTED WIRING TO THE PRESSURE SWITCH WILL RESULT IN DAMAGE TO THE PRESSURE SWITCH. WARRANTY IN SUCH CASE IS NULL AND VOID !

NomaDrive 50 mm switches or "Zero" switches

In conjunction with Optical Safety Edge and other safety devices which are used to Reset the Nomadrive system, a 50 mm Limit Switch is used.

Part number : 96-6025-96 or 96-6020-94 or similar switch

The electrical connection is made according to the schematic diagram as mentioned in the Door Manual supplied with every Nomadrive System.



Furthermore, after installing the 50 mm Limit Switch, the Nomadrive system should be programmed for the use of the Limit Switch.

Therefore the Service instrument must be connected to the J2 connector of the Nomadrive CPU and the "OPTION" mode must be selected.

The following OPTIONS will be displayed : PH.C.1-FUNCTION
-OPEN
-D - L (door line)
-SE (safety edge)
-STS (touchless safety edge)

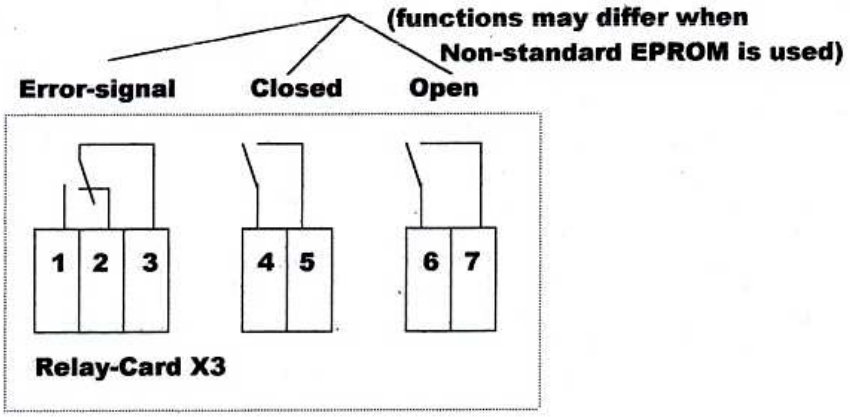
Select : SE. After that dis-connect the service instrument.

The door can be tested by opening the door with the push-button, and closing the door with the push-button. The door should stop and reset the Nomadrive CPU. (look at counter, it should display "0")

Relay Card Error status relay information

The Nomadrive system has an optional Relay-Card to be used as interface between the CPU and external equipment.

Connections:



Part number : 08-8731-00

The Relay-Card gives the following signals:

Error-signal when to following errors occur (also displayed on the screen):

- **1 , 2 , 3 , 4 , 5 , 6 , 7 , 8 , 9 , 10 , 11 , 13 , 14 , 15 , 16 , 18 , 19 , 20 , 21**

The following errors do NOT give an error signal on the Relay-Card:

- **12 , 17**

The signal is available on contacts 1 (NO) , 2 (NC) , 3 (Common)

The following Status signal are available :

- **Door completely CLOSED on contact 4 (NO) , 5 (Common)**
- **Door completely OPEN on contact 6 (NO) , 7 (Common)**

All contacts are rated max. 240 V and are so-called "dry contacts"

Nomafa is not responsible for damage to equipment NOT supplied by Nomafa AB, while being hooked-up to the Relay-Card.

Error codes

At error the error code is displayed on the panel display and is attended to as follows:

01 Overload in the frequency converter
(Reset by new open/close impulse)

02 Motion error / pulse error
(Close the door complete with hold to run)

03 Direction error
(Close the door complete with hold to run)

04 No test impulse from safety edge
(Close the door complete with hold to run)

05 Thermostat in motor released
(Close the door complete with hold to run)

06 Emergency stop relay error
(Close the door complete with hold to run)

07 Emergency stop or crank switch effected
(Reset emergency stop)

08 Memory error
(Close the door complete with hold to run)

09 Opened five times with safety edge
(Reset by new open/close impulse)

10 Error on safety edge
(Close the door complete with hold to run)

11 Error on photocell in door-line
(Close the door complete with hold to run)

12 Opened with safety edge
(Reset by new open/close impulse)

13 No supply voltage to the input
(Run the door by hold to run)

14 Interlocked door is not closed
(Close opposite lock door or open present door with hold to run)

15 Fixing brake released
(Reset fixing brake)

16 Incorrect power supply
(Check the power supply / correct voltage)

17 Opened by photocell in door-line
(Make sure that the door opening is clear from objects)

18 Overload hold to run
(Wait one minute. Close door completely with hold to run)

19 Bottom beam released
(Restore the bottom beam. Close door completely with hold to run)

20 Control system overheated
(Move electric cabinet or mount a ventilator fan)

21 No power supply to the frequency converter
(Contact your local Nomafa supplier for changing the control system)

Default values : SETPOINT

4. Motor	**1.1 kW**
5. Set Speed Up	**50 Hz**
6. Set Slow Speed Up	**10 Hz**
7. Set Speed Down	**50 Hz**
8. Set Slow Speed Down	**10 Hz**
9. Set Acceleration Time	**3**
10. Set Deceleration Time	**3**
11. Set Top Position Distance	**1500**
12. Set Nearly Top Constant	**7**
13. Set Nearly Bottom Position	**100**

Fuses

F1 = 5 VDC Supply (inside CPU)
F3 = 24 VDC Supply (on Terminal Card X1)
+ Inputs (11-18 & 21-35 & 38-39 & 44)
24 VAC Supply (19 a+b)
F4 = 200 - 240 VAC Supply

Fuses F1 + F3 → 1 Amp. (5 x 20 mm, glass)
F4 → 2 Amp. (5 x 20 mm, glass)

QM1 = Circuit breaker 10 Amp. (or 16 A. for 1.5kW)

ERROR CODES FOR NomaDrive system

ERROR CODES : Noma*Drive*

1. **Overload in the Frequency synthesizer**
   - If possible to reset by Power Off / On, Excessive Overload occurred.
   - Under-Voltage occurred if reset is still possible by Power Off / On. Adjust inside CPU Voltage jumper.
   - If reset by Power Off / On is NOT possible, replace CPU. **Fatal error.**
2. **Movement or pulse counter error**
   - Hold to run mode, Close when error is attended to.
   - Optical encoder error. Check the encoder and the cabling between Controlbox and Encoder Check if LED's on CPU (channel A & B) light up when motor is turned, together with the LED's on the encoder board. If not refer to the Nomadrive Manual from Nomafa AB, Sweden.
   - Optical encoder counted in-correctly, because the zero-pulse from the safety edge failed
   - Motor error, check motor
   - Motor runs in the wrong direction. Check motor phase wiring and encoder wiring on terminal card
   - Wrong type of motor programmed or installed for the door
   - Motor is not **DELTA** connected
   - Mechanical movement of door is partly or total obstructed. Check for mechanical damage and bad conditions of steel wires, guide rails , bearings etc.
   - Transmission error. Motor runs but transmission is failing to move door drum
3. **Direction error**
   - Hold to run mode. Close door when error has been attended to.
   - Check motor phase wiring at terminal card.
   - Check if counting on the display adds when moving door up-wards, and decreases if moved down-wards
4. **No test impulse from safety edge - at pneumatic safety edge**
   - Hold to run mode. Close door when error has been attended to.

   See additional leaflet 04 error
5. **Motor temperature error**
   - Check motor and mechanical parts of door for heavy mechanical resistance while in motion
   - Check if correct motor size has been installed and programmed
6. **Emergency stop relay error**
   - Hold to run mode , close the door
   - If error remains even after reset by Power Off / On, replace CPU. **Fatal error.**
7. **Emergency stop or crank switch is activated**
   - Reset Emergency switch or Crank switch
   - Check that they are connected or bridged at the Terminal card.
   - Try to open again
8. **Memory error**
   - EEprom does not work. Hold to run, close door. If fault remains, change CPU. **Fatal error.**
   - Power supply was interrupted while door was in motion
9. **Opened 5 times with safety edge**
   - Check safety edge and the cause of interruption. Check for anything under the door.
   - Check coily cable or air-tube connection if no external cause can be found.
   - The 50 mm limit switch (with optical safety edge) is out of order.
   - Optical encoder counted incorrectly. Hold to run mode, close door.
10. **Safety edge error (Pneumatic safety edge)**
    - Safety edge affected. Check switch, cabling. Check connection to Terminal card.
    - Safety edge (**STS**) affected. Check photocell diodes and cabling. Adjust photocell card potentiometers for optimum adjustment, till LED's light up. Check alignment of **STS** photocells.

11. Photocell in door line error
- Photocell in door line affected, check any obstruction and miss-alignment
- Clean photocell heads if necessary
- Check cabling, and connections
- Check eventual Radars for vibration, and correct by re-inforcing brackets.
- Check eventual photocell cards

12. Opened with safety edge
- Door was opened by safety edge during last opening
- SW 1:1 is incorrectly adjusted on the CPU. Refer to manual Nomadrive from Nomafa AB, Sweden

13. No supply Voltage to the inputs
- Fuse F3 blown
- Motion transformer overload. Check 19 a+b & 20 a+b for excess current. Total current from 19 + 20 should not exceed 1 Amp. (Fuse F3 = 1 Amp.)
- See separate leaflet on Error 13
- If no cure is possible, replace CPU. **Fatal error.**

14. Interlocked door is not closed
- SW1:3 on CPU is 0, if the door should not be interlocked
- Check that interlocked door is closed
- If fault persists, change CPU. **Fatal error.**

15. Fixing brake released
- Manual control mode. Close the door when error is remedied.
- See separate leaflet about error 15
- Check mechanical parts and micro-switch of brake release mechanism
- Refer to the separate test procedure of the CPU, if no other fault are found.

16. Incorrect Power supply
- Mains current too low or too high. Adjust Voltage jumper in CPU. See leaflet.
- Check mains current with Digital Voltmeter. (should be within 10+ or 10-% of recommended Voltage of Nomadrive system.

17. Opened with photocell in door line
- Photocell miss-aligned during last door closing
- Obstacle interrupts the photocell beam

18. Overload hold-to-run mode
- Door has been run too often with manual controls
- Wait a few minutes and try again after cooling off.
- Reset by closing the door completely

19. Bottom beam released (only by break-away systems)
- Lower beam has been moved from its position
- Reset lower beam and close door using manual controls

20. Control system overheated
- Electrical cabinet too warm. Install at cooler place. (see spec. sheets Nomadrive)

21. No power supply to frequency synthesizer
- Power relay or rectifier error. Change CPU. **Fatal error.**

For more detailed info, refer to Nomadrive manual Nomafa Sweden, or contact nearest Nomafa Service center.

Error 04 check

(No test-impulse from pneumatic safety edge)

1. Check led nr. 5 on CPU for going ON/OFF/ON when the safety edge hits the floor. (it should be every single time when safety edge hits the floor, when it misses even one (1) time, error 04 occurs)

2. Check wiring between 17 a&b and the pressure switch, and also the connection to "20 a" if an electronic pressure switch is used

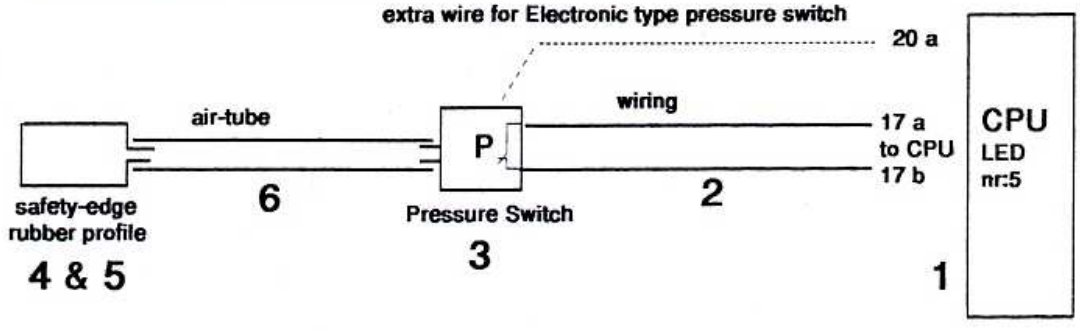
3. Check the adjustment of the pressure switch (non-electronic type) and adjust both the "plastic" screw (membrane), as well as the "metal" screw (air leakage valve). See maintenance manual.

4. Check the position of the safety edge, if the rubber profile is really hitting the floor over the full width of the door.

5. In case of an un-balanced door (2/300 series), check that the bottom beam is really parallel with the floor surface. If not, compensate with rubber blocks glued onto the rubber profile inside the yellow fabric.

6. Check the air-tubes for leakage (even small ones) or bend in the tubes. Look out for air-flow restrictions, as well as "open" airflow conditions. Test by blowing into the begin of the whole tube towards the safety edge rubber profile. It should be closed and one should feel the rubber profile expanding when one blows into the tube. (L-connectors can trick the whole picture, be aware of them)

7. If the fault persists, change the CPU. If this does not help, it surely is one of the above noted failures.



Error 13 check

ERROR 13 : Cause 1
Fuse F3 has been blown, due to a short circuit in the 24 Volt accessories connected to the terminal card connections or the wiring of those accessories.

Cause 2
Accessories connected to terminal 19 a+b (24 VAC)
or
Accessories connected to terminal 20 a+b (24 VDC)

are consuming to much current > 0.8 Amp. together on one or both connections. This is not enough to blow the fuse F3, but the error system is already cutting the 24 Volt supply and generating the ERROR 13 as pre-caution for damage to the Nomadrive system.

REMEDY : Disconnect all wiring from terminal 19 a+b and 20 a+b
and measure the Voltage on 19 a+b (24 Volt AC)
and measure the Voltage on 20 a+b (24 Volt DC)

If Fuse F3 was blown, put new Fuse 1 A (never more) in holder F3

Test the error messages, it should be OK by now.

Put back the connection 19 a+b, and measure the current with a milli-amp. meter in series with 19 a. It should not exceed the maximum of 800 mA, or if both terminals 19 and 20 are used, it should not exceed for both the the terminals max. 800 mA together.

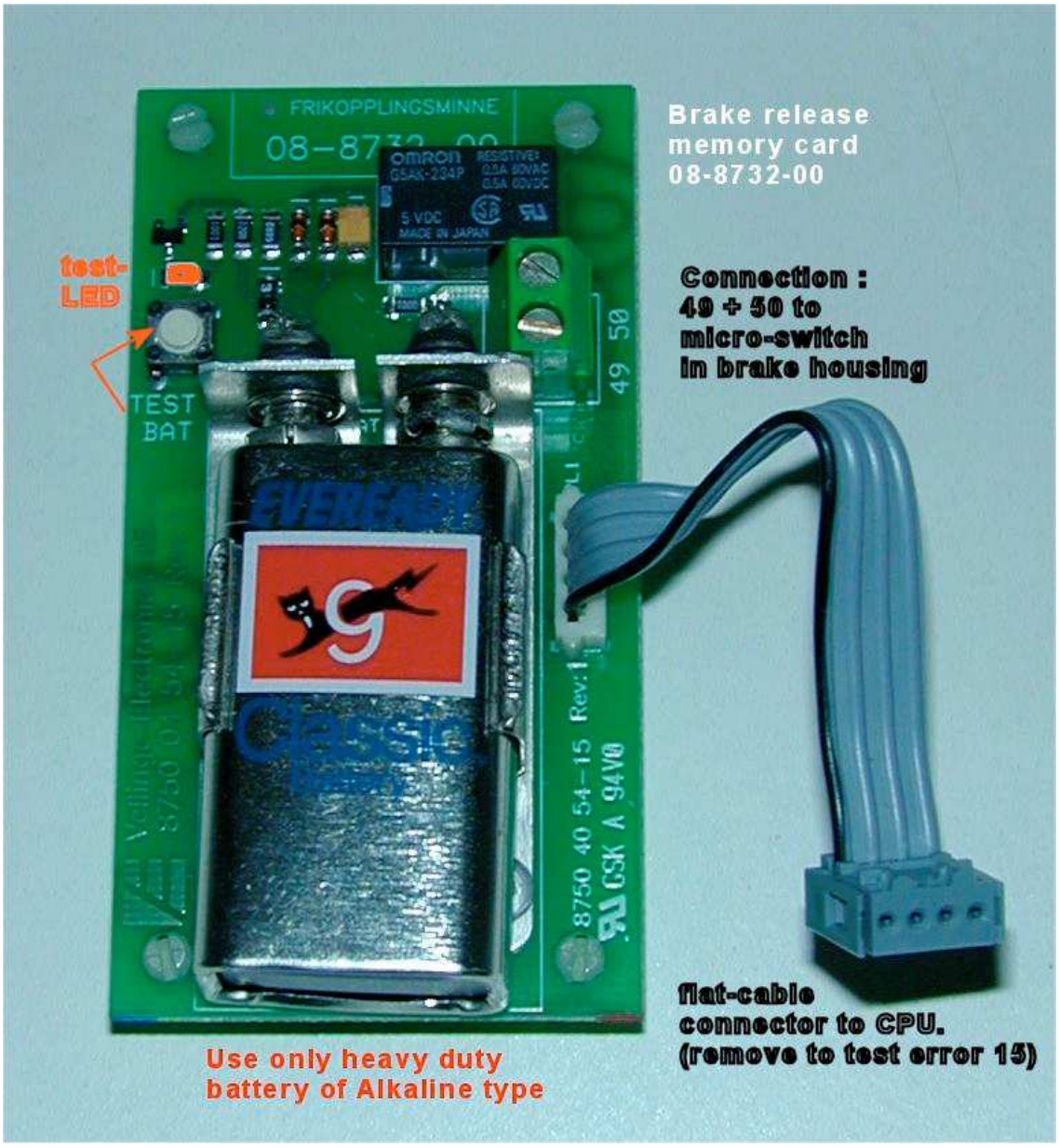
Repeat the same for terminal 20 as described above.

Disconnect any accessory which consumes too much and replace those. (suspect : Radar /Radio /Switches /etc)

CHECK CONNECTOR J-9 on the Terminal card for good contact. Use Voltmeter !

THE "LED NR. 3" SHOULD BE "ON" WHEN CPU IS POWERED "ON".

Brake Release Memory Card



The memory card registers the fact that the brake is released while the power to the control system has been cut-off. Error 15 shows up on the LCD display. Engage brake again and start door with Hold-To-Run.
During service inspections always check battery and proper operation of Micro-Switch in brake housing.

Error 15 check

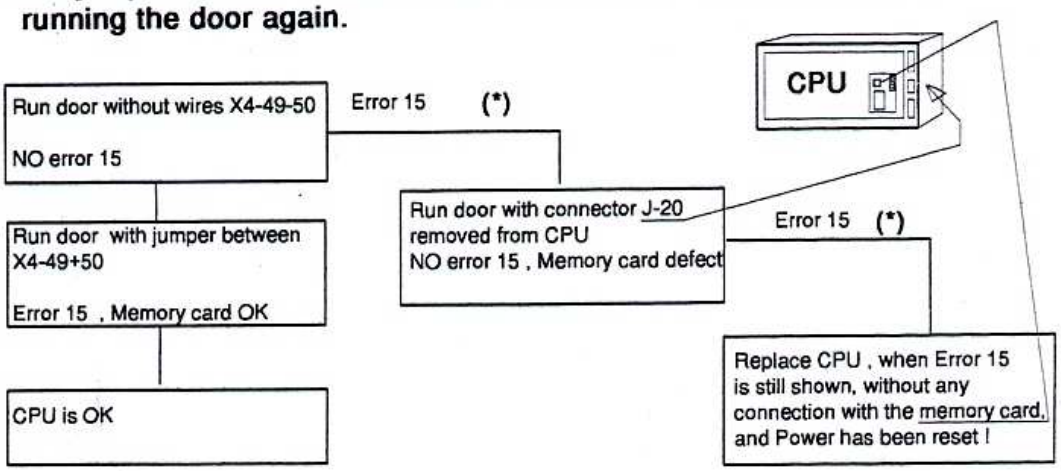
To determine whether the CPU is defective or the brake and it's micro-switch or wiring :

0. Switch off the power with the main-switch

1. Remove both wires from terminal X4-49+50 (brake memory release)

2. Switch Power ON and run door without the wires. There should NOT be any error 15

3. Put jumper between X4-49+50 and look if error 15 shows up after running the door again.



Between each step during the test, marked with (*), reset power of the CPU by switching Main-Switch Off-On.

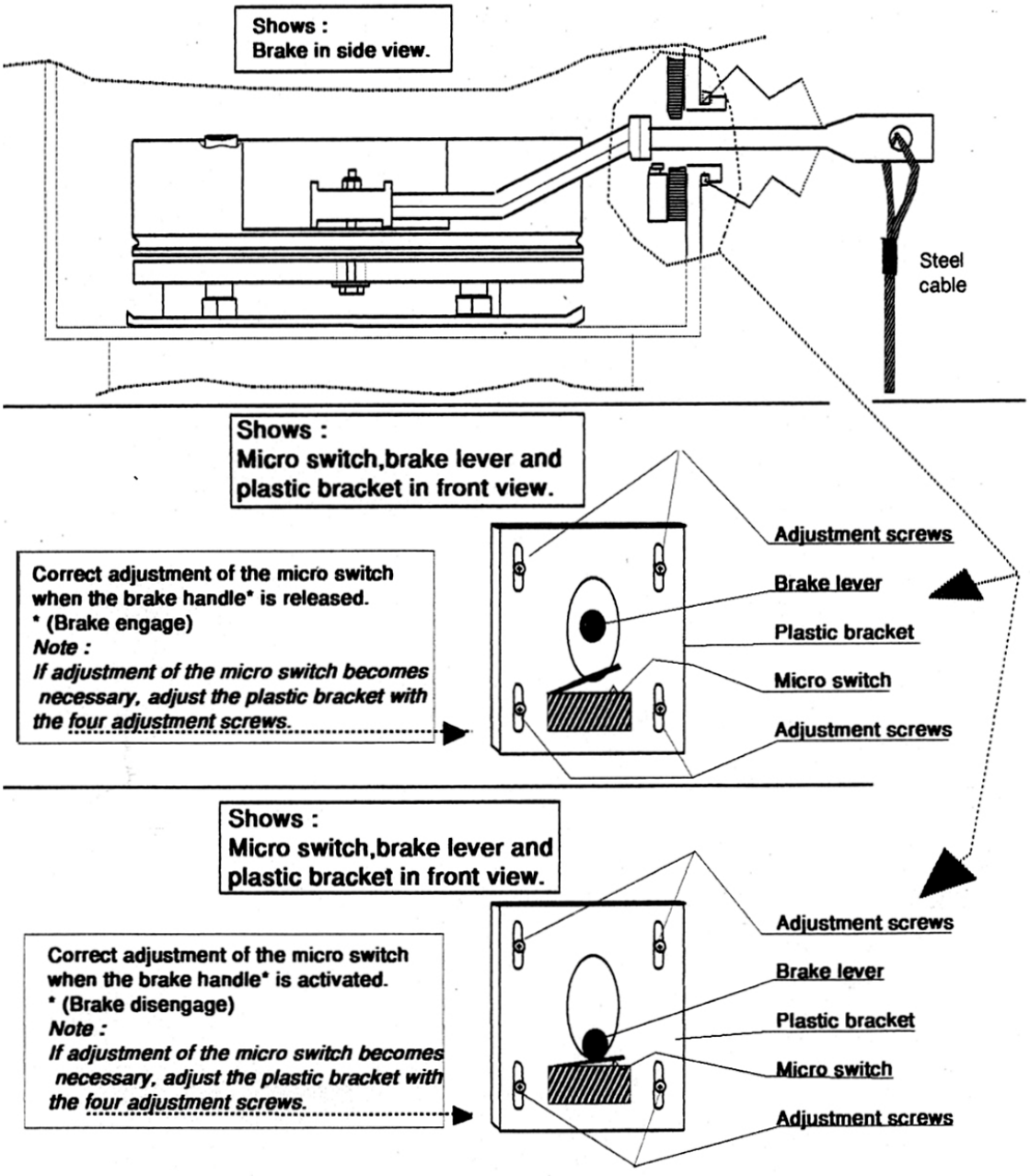
Every other Error 15 cause is NOT related to the CPU !!!

Other causes:
1. **Faulty micro-switch (test with Ohm-meter + initiate switch)**
2. **Faulty wiring (broken electrical cable or similar)**
3. **Wrongly adjusted micro-switch (should be CLOSED when brake is released by handle)**
4. **Release handle or steelcable is defect or wrongly adjusted**

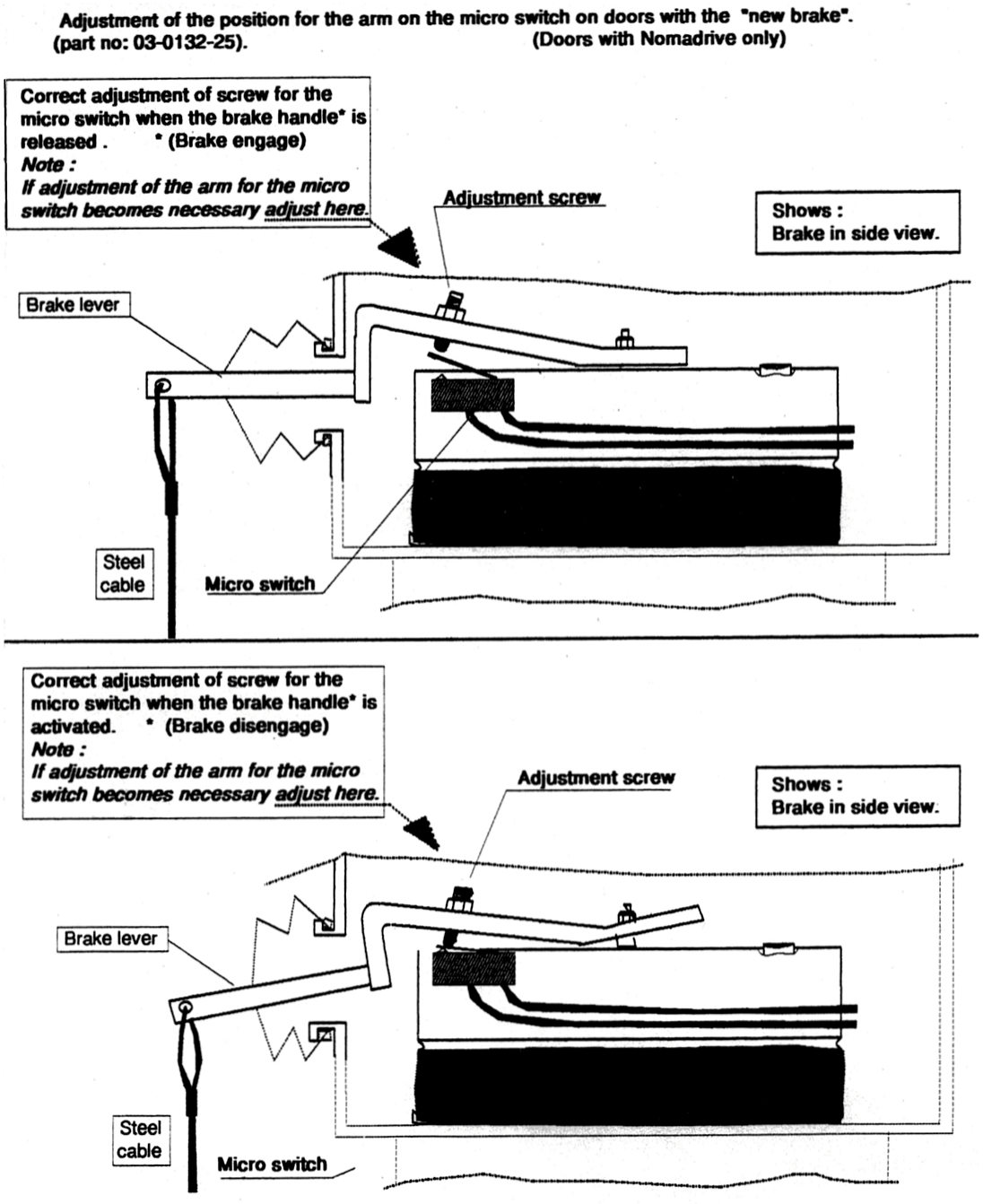
TEST IF BATTERY OF MEMORY CARD IS OK --> PUSH TEST-BUTTON !!!

OLD BRAKE ADJUSTMENTS

Adjustment of position for the micro switch on doors with the "old brake".
(part no: 03-1030-25) (Doors with Nomadrive only)



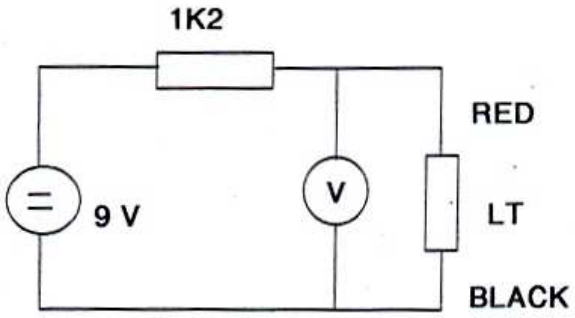
NEW BRAKE ADJUSTMENTS:



Field test Telco Photocells

The following tests can be used to diagnose whether a "Transmitter LT" or a "Receiver LR" is defective:

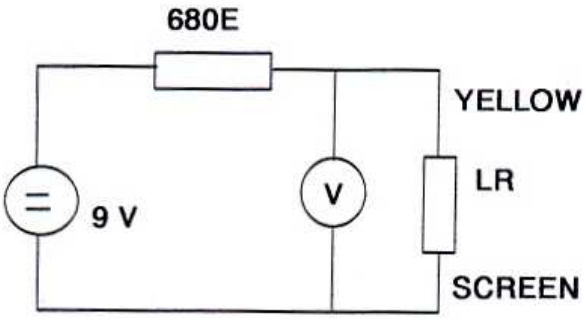
1.



TRANSMITTER: LT

Normally Voltage is between 1-1.5 V

2.



RECEIVER: LR

Normally Voltage is between 3-4 V

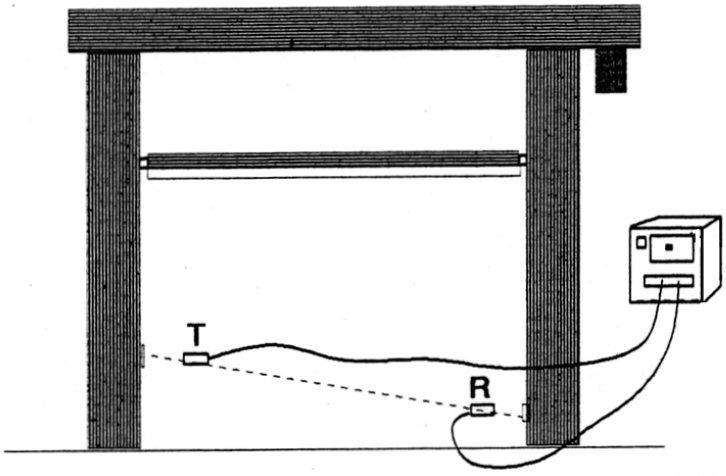
The in picture 1 & 2 given simple tests are meant to test sensors in the field in an easy way. (make notes of values in practice, and gain experience)

This method is indicating - Short circuit
- Broken Connections

It does not indicate complex component failures.

(a normal Volt meter is used : R(meter) > 1 meg.Ohm/volt minimal)

TESTING OF PHOTO-CELLS IN THE FIELD



Quick test : Use a spare set of Telco Photo-cell LT and LR.

1. **Connect "instead of" an existing TL or LR and hold it for the opposite original LT or LR.**
2. **Check if the LED (nr. 6 from the top) is going "ON-OFF".**
3. **Replace the defective one.**
4. **CHECK IF SWITCH 1.4 IS "ON"**

Electronical test :

1. **Interrupt line "RED" (37a) and put mA meter in series and measure the current. (should be constant) (test with good one first, and compare)**
2. **Interrupt line "YELLOW" (37b) and put mA meter in series and measure the current. If beam is interrupted, current should change. (test with good one and compare)**

   FOR THIS TEST MAKE NOTES ABOUT THE VALUES, TO LEARN !!!

NOMADRIVE EPROM IMPROVEMENTS in Eprom 9910xx

Hold-To-Run (HTR) function):

- With switch 6 in ON position it is not possible to run the door over programmed top position in HTR-mode. => Err 18.
- With switch 6 in ON position it is not possible to reset the zero position above 20 pulses.

Not installed

- When powering up, the CPU automatically writes install in the memory.

Reduced Height:

- Improved function when activating open to full height when decelerate to reduced height.

100% Interlock:

- When interlock function is selected, the door is delayed, but relay door closed is deactivated directly. This makes it impossible to open two interlocked doors even if they are activated to open simultaneously.

200/300/400/600/700 in the same Eprom:

- 320 are now included in the standard Eprom

New option El. safety edge:

- When a electrical safety edge is used (8.2 kOhm) it can be supervised by the Nomadrive. This selection is not depending on door type. NOTE if a external module like a Bircher amplifier ESR 32 is used, this option shall not be selected.

New option Input 33:

- It is possible to control if Safety input 33 shall be Normally closed (standard for 300) or Normally open (standard on other doors).

New feature, input 35:

- Brake away. Normally only used on 300 door type.

Reduced closing speed, 300 and El. safety edge:

- When a electrical safety edge is used (8.2 kOhm) on door type 200/300 the closing speed is reduced from 28 Hz to 25 Hz.

Bug correction:

- After stopping a closing door, the doorline photocell opened the door.

Display of opening/closing time:

- The display of Opening/closing time is always set to zero (calculation from door closed to door open and vice versa).

Two opening-heights on 200/300:

- Handle “Two opening-heights” on 200/300 is not possible to use in the standard program.

Note:

The above mentioned improvements have been made permanently in all new standard EPROM programs from Dec. 1999 onwards.

The improvements are NOT available for the older 4000 and 3000 CPU’s still In use on older doors.

Note 2003-1 :

Latest versions EPROMS : standard programs **021206** superseded by:

- **030602 for 420 / 660 doors up to 6000mm wide (conical drums)**
- **050702 for 660 doors from 6001 mm wide onwards (cylindrical drums) (22 Hz lower top-frequency)**
- **030131 for 330 doors with Zero-trigger circuitry**

The new EPROMS are only valid for serial “5000” CPU’s

Space for your own notes:

Interlock system NomaDrive

INTERLOCK FUNCTION BETWEEN TWO DOORS WITH NOMADRIVE

It is possible to use two doors together in an "Interlocked" mode, while using "Opening with automatic closing" mode.

- When door 1 is open, the relays output (Door 1 NOT CLOSED) of door 1 disables starting impulses for door 2.

- When door 2 is open, the relays output (Door 2 NOT CLOSED) of door 2 disables starting impulses for door 1.

In this case, dip-switch nr. 3 on the CPU enables this function, if so desired.

Use of the normal push-button "Up-Down" overrides this interlock feature.

IMPORTANT :

This configuration makes it necessary to have each door equipped with all initiators which will allow each door to be operated individually as single door.

Loops and/or pull-switches and/or pushbuttons have to be installed for each individual door at both sides of the doors.

Use of radar instead of induction loops is not advisable.

On door A :

Output relays : 43a-43b (door A gives info to Door B)
Input relays : 44a-44b (door A receives input from Door B that Door B is closed)

On door B :

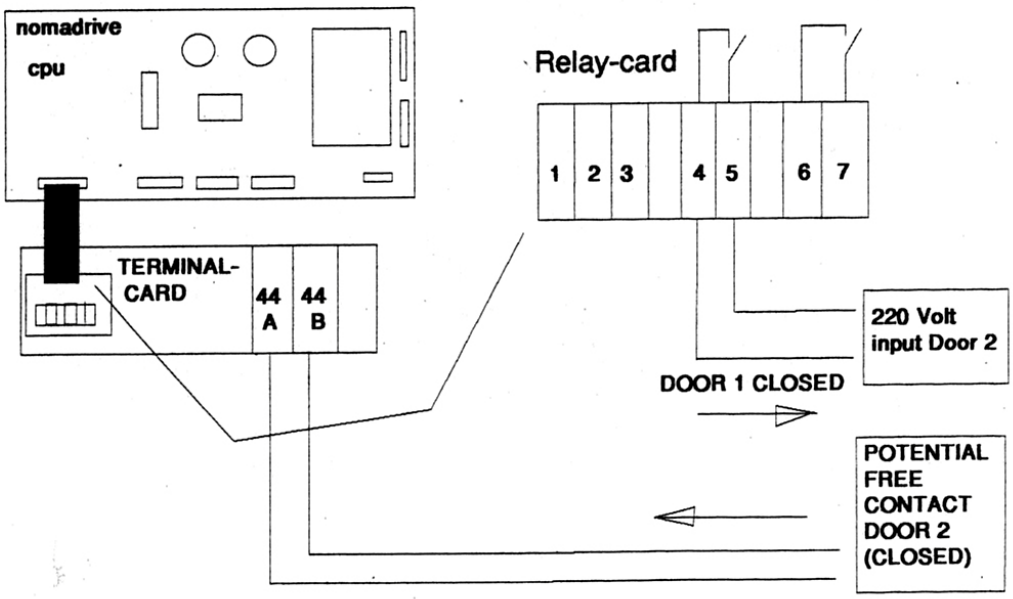
Output relays : 43a-43b (door B gives info to Door A)
Input relays : 44a-44b (door B receives input from Door A)

A four-core cable should run between door A and B.
This cable has to be provided by the Installer.

INTERLOCK BETWEEN NOMADRIVE & DOOR OF OTHER BRAND (220 V)

To interlock a Nomadrive Door with a door of an other brand, the Nomadrive system has to be extended with a Relay Card. (08-8731-00)

The connections have to be made according to this schematic diagram:



NOMAFA suggests strongly to use 24 Volt control Voltage between the 2 doors being interlocked.
Door 2 (non-Nomafa) should have a Relay output (potential free) which contact is Closed if door 2 is closed. (connect to 44 a+b Nomadrive)
Door 2 receives from the Nomadrive door a closed contact of door 1 is Closed. (Potential free contact) [4 & 5 on Relay card]

Be aware of the fact that door 2 should have its own initiators.

The way of operating is described in Nomafa Information about interlock.

APPLICATION OF THIS CIRCUITRY IS FOR RISK OF END-USER.

NomaDrive Airlock system (options)

OPTION 50.108 : Airlock function, Control box A - Nomadrive (including)

- Photocell Card (interface)
- Photocell (receiver / transmitter)
- Brackets for Photocell
- Summer / Winter Switch
- Special Airlock program in EPROM
- Relay card

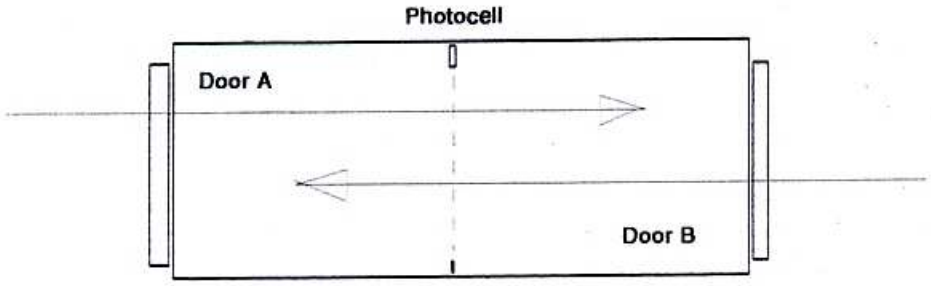
OPTION 50.109 : Airlock function, Conrol box B - Nomadrive (including)

- Summer / Winter Switch
- Special Airlock program in EPROM
- Relay card

The normal standard Photocell in door line is not included in this Option, but is part of the standard specification of the door.

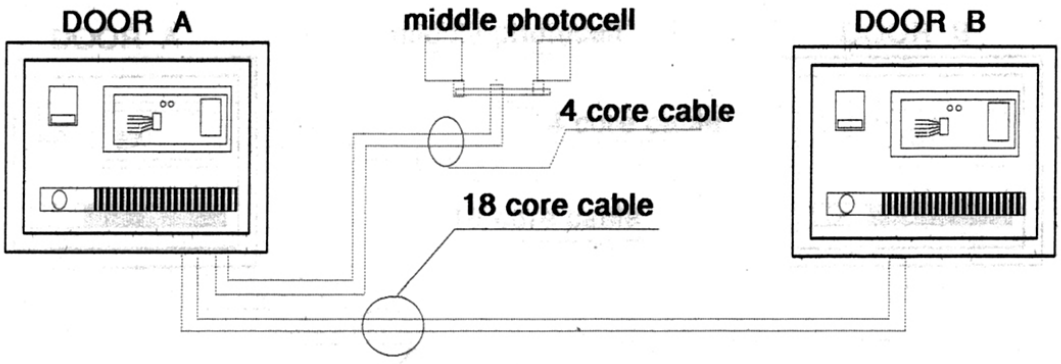
Note : The special Photo cell should be mounted in the middle of the airlock :

Note 2 : For Functional description see leaflet : Functions Airlock



FOR ALL STANDARD ELECTRICAL CONNECTIONS REFER TO YOUR NOMADRIVE MANUAL SUPPLIED WITH YOUR DOOR

TECHNICAL INFORMATION : AIRLOCK NOMADRIVE CONNECTIONS



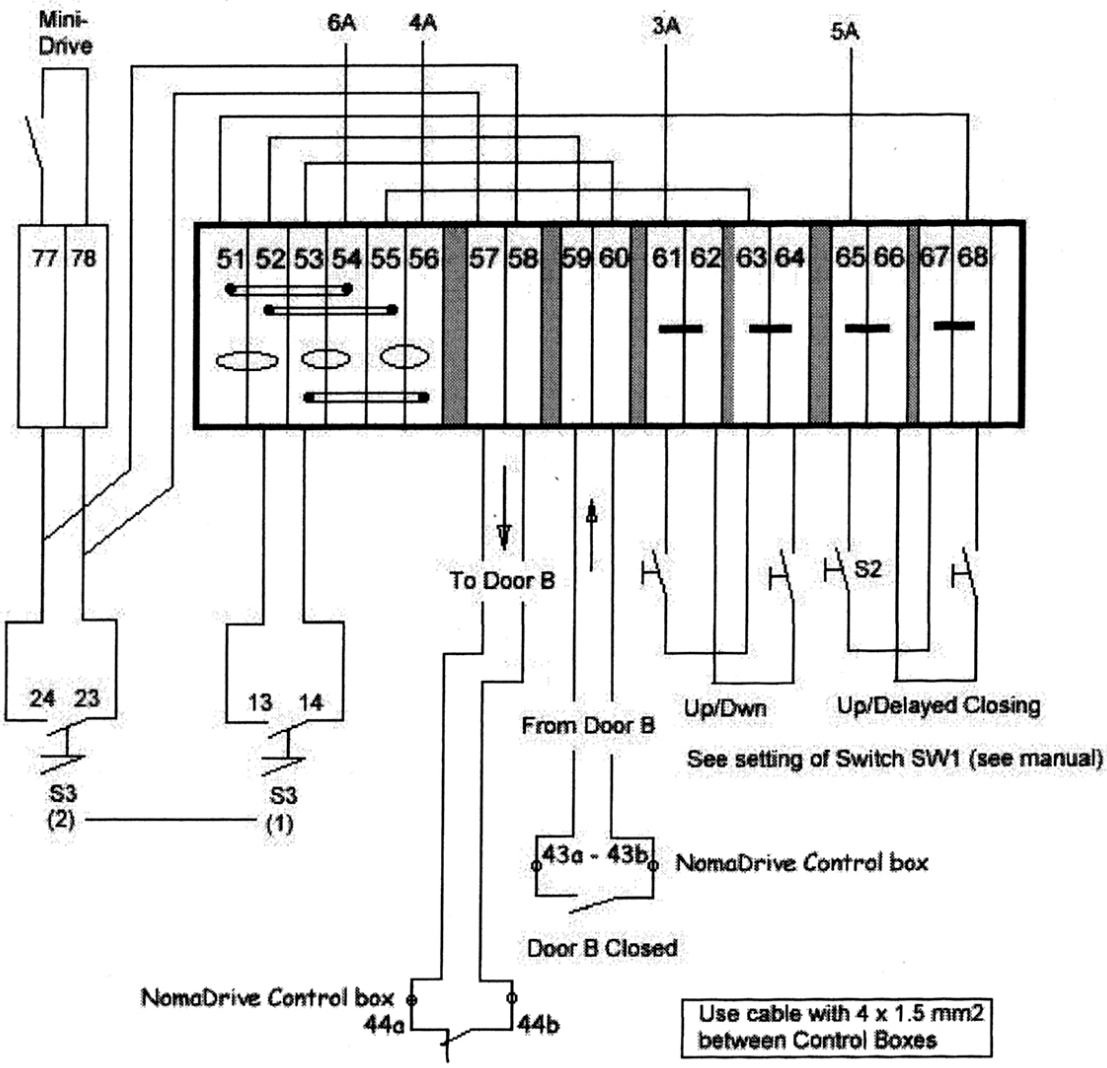
Core nr:	Function	Terminal Card A		Terminal Card B
1	Door A closed (output)	X1.43.A	→	X1.44.A (input)
2		X1.43.B	→	X1.44.B
3	Door B closed (input)	X1.44.A	←	X1.43.A (output)
4		X1.44.B	←	X1.43.B
5	Opening Impulse (output)	X1.42.A	→	X1.18.A (input)
6	To Door B	X1.42.B	→	X1.18.B
7	Opening impulse (input)	X1.18.A	←	X1.42.A (output)
8	To Door A	X1.18.B	←	X1.42.B
9	Door A in (output)	X3.6	→	X1.33.A (input)
10	Summer mode	X3.7	→	X1.33.B
11	Door B in (input)	X1.33.A	←	X3.6 (output)
12	Summer mode	X1.33.B	←	X3.7
13	Photocell in (output)	X3.4	→	X1.34.A (input)
14	Door line Door A	X3.5	→	X1.34.B
15	Photocell in (input)	X1.34.A	←	X3.4 (output)
16	Door line Door B	X1.34.B	←	X3.5
17	Middle Photocell (output)	X3.1	→	X1.35.A (input)
18		X3.3	→	X1.35.B

X1.xx.y are Terminal card connections
X3.xx are Relay Card connections

All Outputs are relay-contacts ! All inputs : Ports !

NOTE: THE ABOVE CONNECTIONS ARE NOT INCLUDING THE STANDARD CONNECTIONS FOR EACH DOOR !!!

Interlock Connections between MiniDrive & NomaDrive systems



This drawing shows Control Box Door A

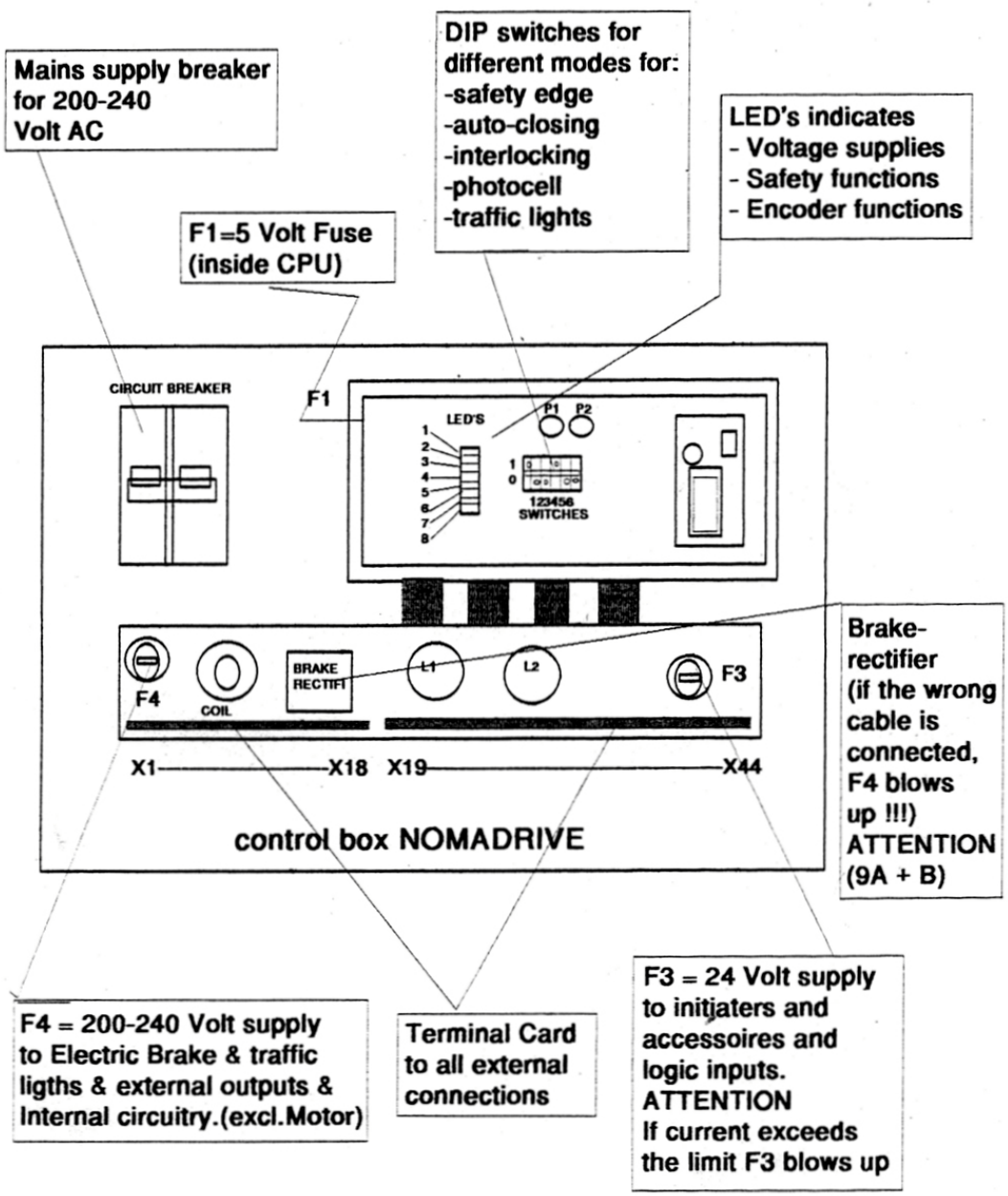
12-09-2000

Legend:

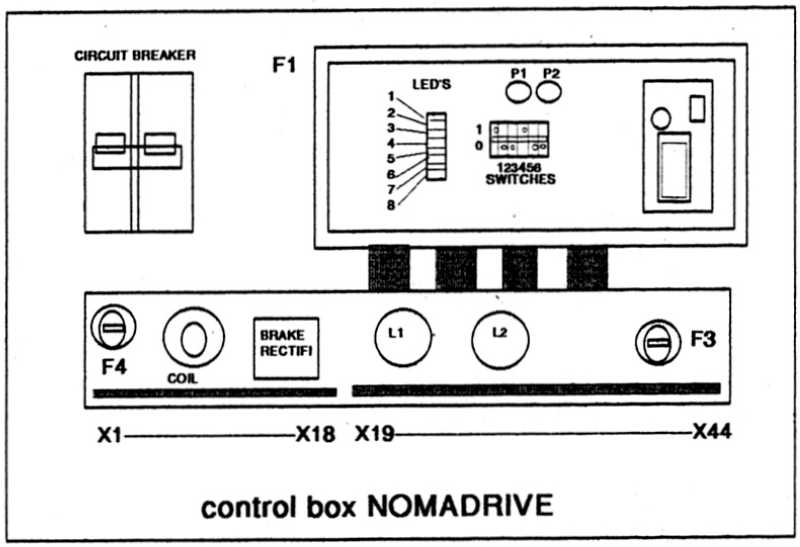
⬭ = Jumper when Opening with Delayed Closing is requested (SW1=ON)

▭ = Jumper when Opening and Closing is without delayed action.(toggle) (SW1= OFF)

TECHNICAL INFORMATION : NOMADRIVE CPU-TEST



TECHNICAL INFORMATION : NOMADRIVE CPU-TEST



control box NOMADRIVE

To test the Nomadrive CPU in an efficient way :

1. Connect Motor	to	1 + 2 + 3 + PE	
2. EL-Brake	to	9a + 9b	(be careful to pick the correct cable !!!!!)
3. Encoder	to	11a + 11b + 12a + 12b	(brown, white, green, yellow) CHECK !!!
4. Brake Release	to	49 + 50	(brake release)
5. Put "jumper"	to	16a + 16b	(thermostat in motor)
6. Put "jumper"	to	13a + 15b	(emergency switches)
7. Put "jumper"	to	17a + 17b	(pressure switch)
8. Put "jumper"	to	23a + 24b	(stop switches)

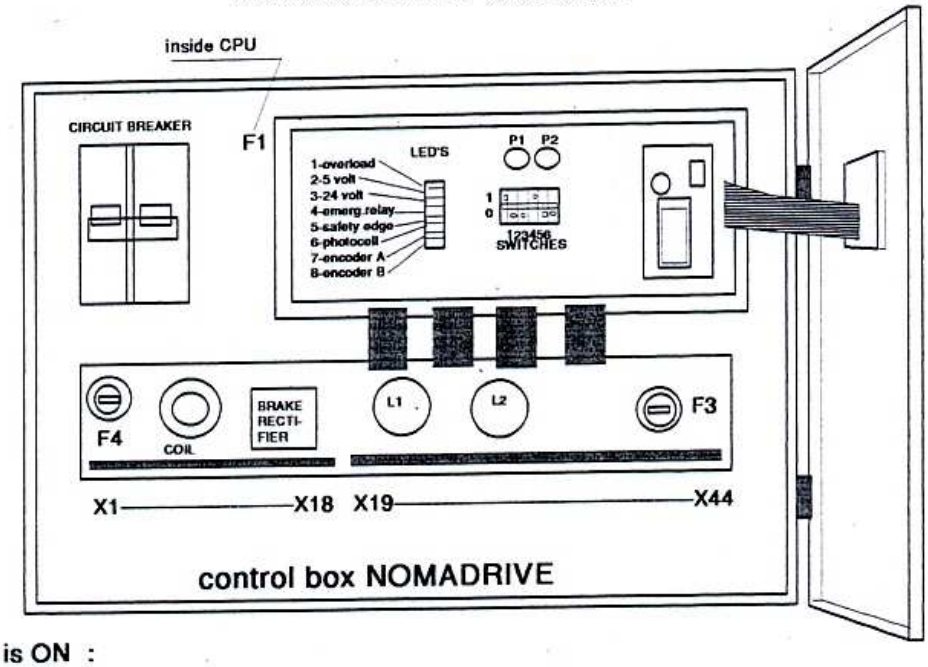
REMOVE ALL OTHER WIRING !!!

DOOR MUST BE ABLE BY NOW TO RUN WITH :

HOLD TO RUN MODE

SEE PAGE #2

NomaDrive CPU test #2



If Power is ON :

1 - 200-240 VAC on connection 4 + 5 (Volt meter)

2 - LED 's	1 (Overload)	OFF	(check Motor / F4)
	2 (5 Volt DC)	ON	(check F1)
	3 (24 Volt DC)	ON	(check F3)
	4 (Emergency Relay)	ON	
	5 (Safety Edge)	ON	(always ON during test)
	6 (Photocell)	OFF	(Switch 4 OFF)
	7 (Encoder A)	ON / OFF	(switching when motor turns)
	8 (Encoder B)	ON / OFF	(switching when motor turns)

3 - If F4 blows after starting motor > Check the BRAKE RECTIFIER

4 - Check for 24 VAC on 19a + 19b and for 24 VDC on 20a + 20b (Voltmeter)

If motor runs OK, then replace the "jumpers" step by step with the original accessoires. Start with the "Thermostat" in the motor (16 a + b), then the emergency switches, the Stop switches, the pressure switch of the safety edge, and check after each step if everything is still OK. If one of the steps cause problems, eliminate that problem and proceed in a "step by step" approach to the next step.

NOTE: A "jumper" is a piece of wire, used to "bridge" a connection -->



Technical Data

Control system:

Type	Nomadrive	
Supply voltage	200, 230, 240 VAC ± 10 %	
Max fuse incoming voltage	10 A	
Data outputs:	Motor	200, 230, 240 VAC Short-circuit proof
	Traffic light	200, 230, 240 VAC, 0,5 A
	Brake	190 VDC 0,1 A
	Door open	24 VAC 0,25 A
	Door closed	24 VAC 0,25 A
Supply impuls sensor	24 VAC / 24 VDC } 1,2 A totally	
Data inputs	24 VDC + 30 % 15 mA	
Analogue data input	0 - 10 VDC, max. 20 VDC 0 - 0,7 mA 15 kOhm	
Ambient temperature	-10 °C bis +50 °C	
Storage temperature	-40 °C bis +80 °C	
Insulation class	Class II via opto switch control circuit and transformer	
Noice tested	ESD safe up to 20 kV EMS safe according to VDE 0875 (CISPR 14), VDE 0871	

Motor:

Type	3-phase asynchronous motor	
Effect	1,1 kW	1,5 kW
Rated voltage	220/380 V	220/380 V
Rated current	5,0/2,9 A	7,35/4,25 A
Rotation speed	1350 rpm	910 rpm
Power factor	0.81	0.66
Number of poles	4	6
Protection class	IP 54	IP 54
Flange	B 14	B 14
Standard	IEC 34-1	IEC 34-1
Insulations class	IEC 85 Class B	IEC 85 Class B
Loading-limiting device	Winding thermostat	Winding thermostat

Gear:

Typ	1-step worm gear
Gear ratio	10 : 1
Torque	170 Nm
Max temperature	100 °C

Fixing brake:

Type	Electro-mechanical
Supply voltage	190 + 10 % VDC
Brake torque	14 Nm
Insulation class	IEC 85 Class B
Protection	IP 54
Max ambient temperature	70 °C

Pulse encoder:

Typ	Optic
Supply voltage	24 + 10 % VDC
Protection	IP 54
Max ambient temp.	60 °C

Installation of limit switch for zero position on a 330 door.

Background:
The limit switch are implemented to improve the look on the door blade and to make sure that the control system gets a zero position when the door is in closed position.

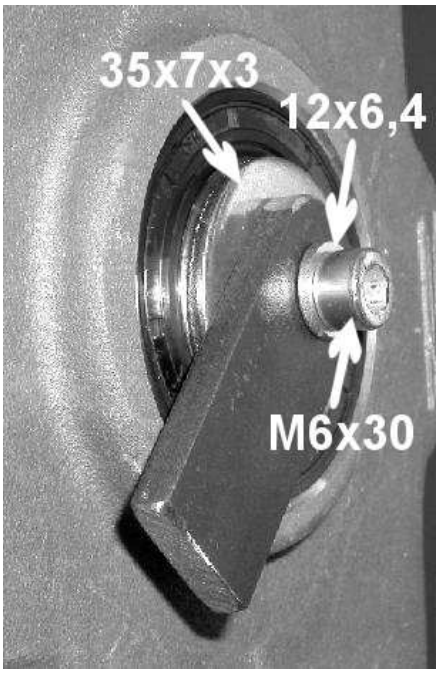
Installation:
Fix the trigger on the drive shaft. There shall be a washer 35x7x3 between the shaft and the trigger. The shall be a washer between the trigger and the screw (MC6S 6x25)
Install the bracket with the screw for the vibration damper and on the other side of the gearbox with a screw MC6S 10x100 12.9 (black). Make sure that the hole in the “centre” of the bracket is in front of the drive shaft – the second hole shall be towards the hall (not the wall). Install the inductive sensor in the “second” hole. Make sure that there is 2mm gap between the sensor and the trigger.
Install the sensor electrical.
Switch the E-prom.

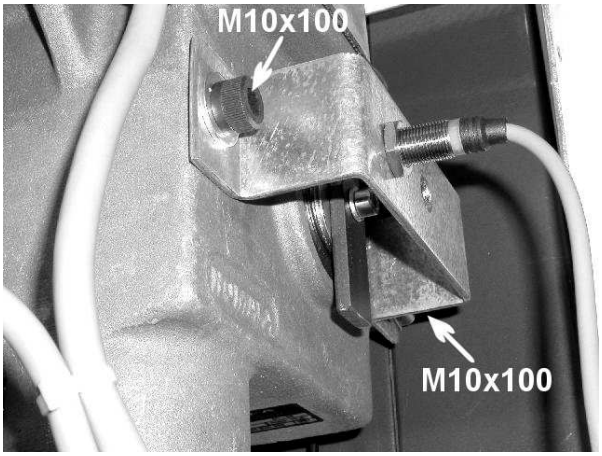
Start up:
Close the door in hold to run mode against the floor
Observe!!! The door must close to the floor and get a zero position from the electrical edge whenever the door is in hold to run mode.
Set the trigger so it gives a signal to the inductive sensor. Run the door one cycle. If needed adjust the trigger till the door stops on the right position.
Go over in hold to run mode again. Run the door down in hold to run mode to the floor and get a zero position. Run the door normal – make sure that the door finds the zero position from the inductive sensor.
Make sure that the screw for the trigger is locked.

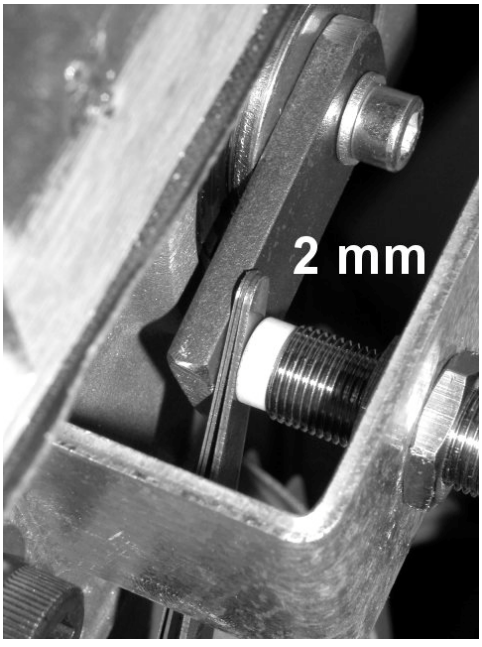
When the zero trigger is working correctly, it generates a pulse shortly **before** the safety edge touches the floor. The zero trigger pulse must be prior to the pulse generated by the safety edge !!! Re-adjust if necessary.

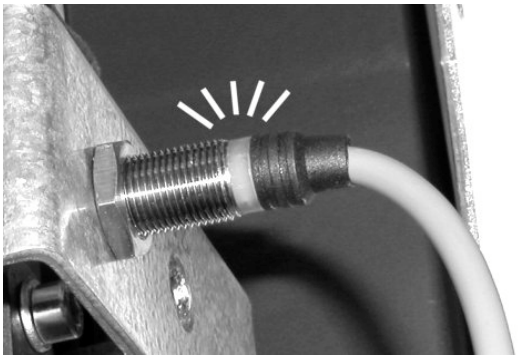
Hand over to customer:
Let the customer know that in case of power failure or other cases when the door comes in hold to run mode must the door close to the floor.

If you have any questions please contact ADS (Asia) in Malaysia. (60-3-7727 0184 ph/fx)


35x7x3
12x6,4
M6x30


M10x100
M10x100


2 mm



Technical Information Pulse Encoder Disc

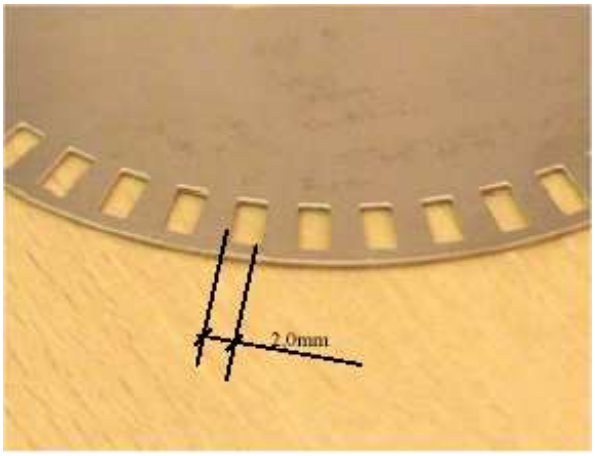
Problem
Sometimes the door passes past the set top position and the bottom beam crash in to the top roll/ top roll cover. This has been known to be a problem on doors where the top motor speed is set to more than 75Hz, for example the RR330 or any other door where the top motor speed is set to more than 75 Hz.
Due to tolerance variations, the problem has occurred randomly, and more seldom also on doors not running faster than 75Hz.

Solution
Change the pulse encoder disc to the new type. If the speed or the door earlier has been set lower to correct the problem of opening too high, the speed can be set back to the desired value. The spare part number for the disc mounted on the plastic centre hub is 3201V0006.
Change of pulse encoder disc is only required where the above mentioned problem has occurred. From 2005-02-18 the new disc is replacing the old type in production.

Where used?
Drive units for Albany Door System doors with control system ACS70, except model RR300/RR320.

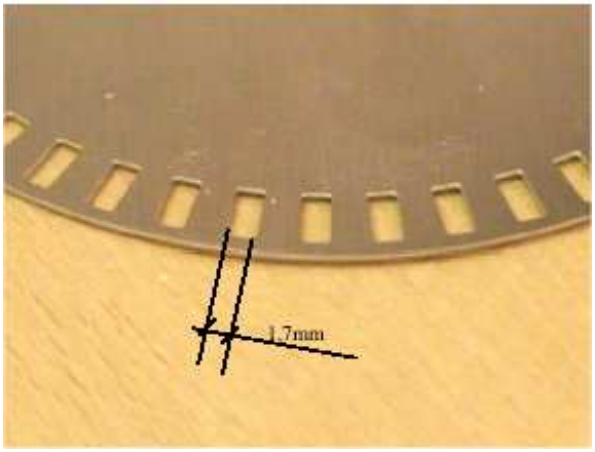
What does it do?
It is used as position sensor for the door, a 2-channel infrared optocoupler counts the holes in the disc, which is mounted on the primary shaft of the gearbox.

How does it look like?
Old type



2,0mm slit width

New type



1.7 mm slit width

Result of the difference

The asymmetric relation between the hole width on the new disc result in a symmetric relation between high and low value of the electric digital signal to the control system, thus preventing problems related to the old disc with the controller not “counting” or sensing the door changing position at high speed.
1,7mm

Contribution:

Anders Stening 2005-02-17 2(2)

Notes:

For help with error finding in safety edge and brake module and pulse counting

(The samples given here are for a 660-SRD-Frigo-H door.
Make sure you refer to the el.diagram of the correct door type when using it for other ACS70 / NomaDrive system equipped doors. The principles are the same, the terminal card numbers may differ !!!)

Trouble shooting for pulse counting and brake function ACS70

(pressostat (red) in RH upper corner of controlbox)

zero pulse (reset) → To CPU via pressostat in Contr. box X1-17A & 17B & 20A
(when safety edge hits the floor in bottom position)

- Check LED nr. 5 on CPU for operation **(safety edge)**
- Use Display on cabinet step 1 for position of door in pulses
- Check air-tube for leaking or bad connection to pressostat
- Check safety edge rubber profile for air leaking (blow on tube for test)

Pulse counting — Door should always start with "0" and then count upwards when opening

(in top of gearbox housing)

Encoder board + disk → To CPU via X1-11A & 12A (channel A and Channel B)
(11B=24 VDC + and 12B= 24 VDC -)

Check LED nr.7 (channel A) and LED nr.8 (channel B) for operation

LED's should light up when moving door slowly by hand,

when brake is released and emergency stop is applied.

not even one pulse should be missed !

Parking brake in gearbox house ← Brake module on terminal card ← CPU

X1-9A & 9B

190 VDC
Brake coil connection
Only powered on during running !!!

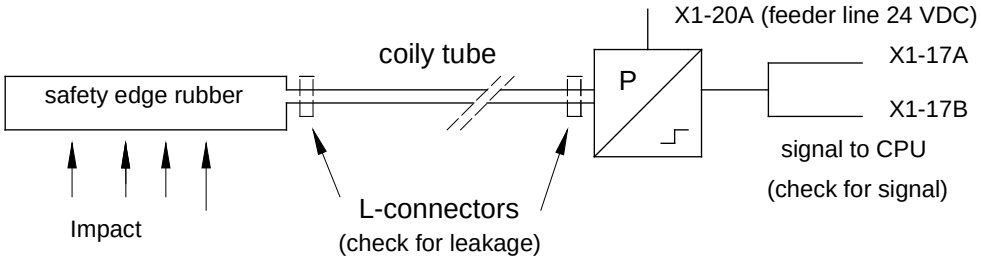
J14-1 & J14-3

Look for LED on module

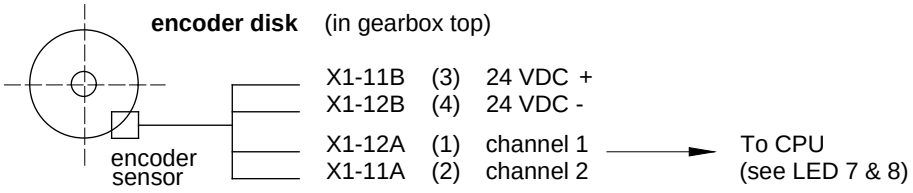
"ON" when brake is electrically released

Check for mechanical malfunctioning
(see page 60 of ACS70 Blue Book)
Free play of brake disk is 0.2 mm when brake is released !

Safety edge on bottom beam



Safety edge system with pressostat in Control Box



Parking brake:

